

**BIOLOGICAL HEALTH RISKS
QUALITY OF LABORATORIES**

EXTERNAL QUALITY ASSESSMENT*

DEFINITIVE GLOBAL REPORT
FLOW CYTOMETRY: LYMPHOCYTE SUBSET ANALYSIS
SURVEY 2026/1

* RD 03/12/1999

Sciensano/Flow cytometry/97/E

Biological health risks
Quality of laboratories
Rue Juliette Wytsmanstraat 14
1050 Brussels | Belgium

www.sciensano.be

COMMITTEE OF EXPERTS

Sciensano					
Secretariat		PHONE:	02/642.55.22	FAX:	02/642.56.45
		E-mail:	QL_secretariat@sciensano.be		
Dr. Bouacida L.	Coordinator	PHONE:	02/642.53.83		
		E-mail:	lobna.bouacida@sciensano.be		
Dr. Vernelen K.	Alternate coordinator	PHONE:	02/642.55.29		
		E-mail:	kris.vernelen@sciensano.be		
Experts	Institute				
Dr Chatelain Bernard	UCL Louvain				
Dr Defour Jean Philippe	CHC Mont Léglia				
Dr Gothot André	ULG Liège				
Dr Hofmans Mattias	UZ Gent				
Dr Keutgens Aurore	ULG Liège				
Dr Kornreich Anne	Grand Hopital de Charleroi				
Dr Mullier Francois	UCL Louvain				
Dr Nguyen Vo Thanh Phuong	Centre Hospitalier EpiCURA				
Dr Poutakidou Danai	LHUB-ULB				

A draft version of this report was submitted to the experts EQA Flow Cytometry on: 17/03/2026.
 This report was discussed at the meeting of the Committee of experts EQA Flow Cytometry on: 31/03/2026.

Authorization of the report: by Lobna Bouacida, coordinator

Date of publication: 16/04/2026

All the reports are also available on our webpage:

- NL: <https://www.sciensano.be/nl/kwaliteit-van-laboratoria>
- FR: <https://www.sciensano.be/fr/qualite-des-laboratoires>

TABLE OF CONTENTS

INTERPRETATION OF THE INDIVIDUAL REPORT	4
SAMPLE MATERIAL	7
PARTICIPATION.....	8
RESULTS	9

INTERPRETATION OF THE INDIVIDUAL REPORT

Besides this global report, an individual report is at your disposal via toolkit.

Below you can find information to help you interpreting this report.

The position of your quantitative results is presented on the one hand in comparison with the results from all the participants and on the other hand in comparison with the results of the laboratories using your method.

Following information is provided:

- Your result (R)
- Your method
- Global median (M_G):
central value of the results obtained by all laboratories (all methods together).
- Global standard deviation (SD_G):
measure of the spread of the results obtained by all the laboratories (all methods together).
- Global median of your method (M_M):
central value of the results obtained by the laboratories using your method.
- Standard deviation of your method (SD_M):
measure of the spread of the results obtained by the laboratories using your method.
- The coefficient of variation CV (expressed in %) for all laboratories and for the laboratories using your method:
 $CV_M = (SD_M / M_M) * 100 (\%)$ and $CV_G = (SD_G / M_G) * 100 (\%)$.
- Z score:
difference between your result and the median of your method (expressed as a number of SD): $Z_M = (R - M_M) / SD_M$ and $Z_G = (R - M_G) / SD_G$.
The result is flagged when $|Z_M| > 3$.
- U score:
relative deviation of your result from the median of your method (expressed in %):
 $U_m = ((R - M_M) / M_M) * 100 (\%)$ and $U_G = ((R - M_G) / M_G) * 100 (\%)$.
The result is flagged when $|U_M| > d$, where "d" is a parameter-dependent fixed limit, namely the percentage maximal deviation from the method median.
- A graphical interpretation of the position of your result (R), towards the results of all the participants as well as the results of the participants using your method, based on the method of Tukey, for each parameter and for each analyzed sample.

R : your result

$M_{M/G}$: median

$H_{M/G}$: percentiles 25 en 75

$I_{M/G}$: internal limits ($M \pm 2.7 SD$)

$O_{M/G}$: external limits ($M \pm 4.7 SD$)

The global graph and the one of your method are presented on the same scale, which allows you to compare them. These graphs give you a rough estimation of the position of your result (R) with respect to the medians ($M_{M/G}$).

More information can be found in the brochures available on our website (only in Dutch and French):

Klinische gezondheid | EKE klinische biologie | sciensano.be

- Algemene informatiebrochure EKE
- Statistische methoden gebruikt voor EKE
- Verwerking van gecensureerde waarden

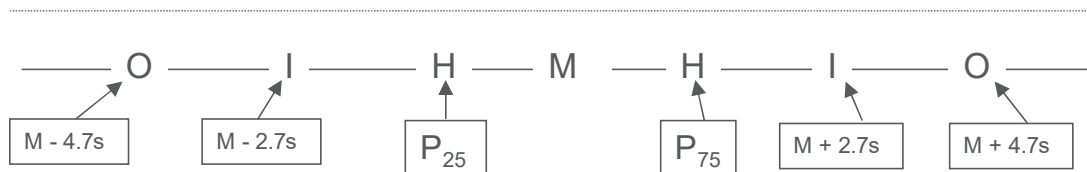
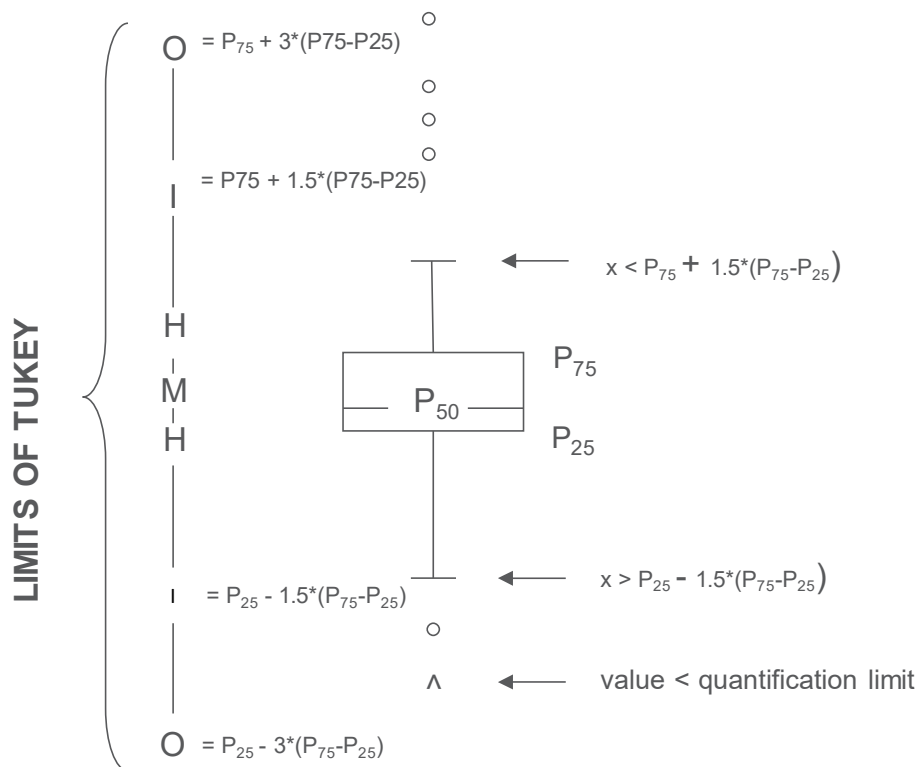
Santé clinique | EEQ biologie clinique | sciensano.be

- Brochure d'information générale EEQ
- Méthodes statistiques appliquées à l'EEQ
- Traitement des valeurs censurées

Graphical representation

Besides the tables with the results a "Box and whisker" plot is added. It contains the following elements for the methods with at least 6 participants:

- a rectangle ranging from percentile 25 (P_{25}) to percentile 75 (P_{75})
- a central line representing the median of the results (P_{50})
- a lower limit showing the smallest value $x > P_{25} - 1.5 * (P_{75} - P_{25})$
- an upper limit representing the largest value $x < P_{75} + 1.5 * (P_{75} - P_{25})$
- all points outside this interval are represented by a dot.



Corresponding limits in case of normal distribution

SAMPLE MATERIAL

Two blood samples (FC/22070 and FC/22071) collected on K2EDTA were sent to the laboratories.

These two samples were collected from two healthy and voluntary blood donors and distributed into aliquots at Sciensano.

The samples were dispatched via BPost 24h, and the laboratories were notified by email about the shipment of the control material (day 0).

Homogeneity was confirmed based on white blood cells determination.

Control analysis on the day of collection and distribution yielded the following results (UZ Brussel):

FC/22070

	%	10 ⁹ /L
Leukocytes		6.42
Lymphocytes	22.1	
CD3⁺ cells	76.1	1.08
CD4⁺CD3⁺ cells	42.3	0.60
CD8⁺CD3⁺ cells	34.6	0.49
CD19⁺ cells	20.5	0.29
NK cells	3.1	0.04
κ % B lymphocytes	65.9	
λ % B lymphocytes	34.1	
κ/λ ratio	1.93	

FC/22071

	%	10 ⁹ /L
Leukocytes		7.51
Lymphocytes	35.6	
CD3⁺ cells	65.4	1.75
CD4⁺CD3⁺ cells	34.5	0.92
CD8⁺CD3⁺ cells	30.0	0.80
CD19⁺ cells	13.4	0.36
NK cells	20.1	0.54
κ % B lymphocytes	63.5	
λ % B lymphocytes	36.5	
κ/λ ratio	1.74	

PARTICIPATION

Forty-nine Belgian clinical laboratories participated in the survey 2026/1 (send-out of blood samples on February 16, 2026 (day 0)).

RESULTS

Regarding sample delivery, 27% of participants received their samples on Day 1 post-dispatch, 67% on Day 2, and 6% on Day 3.

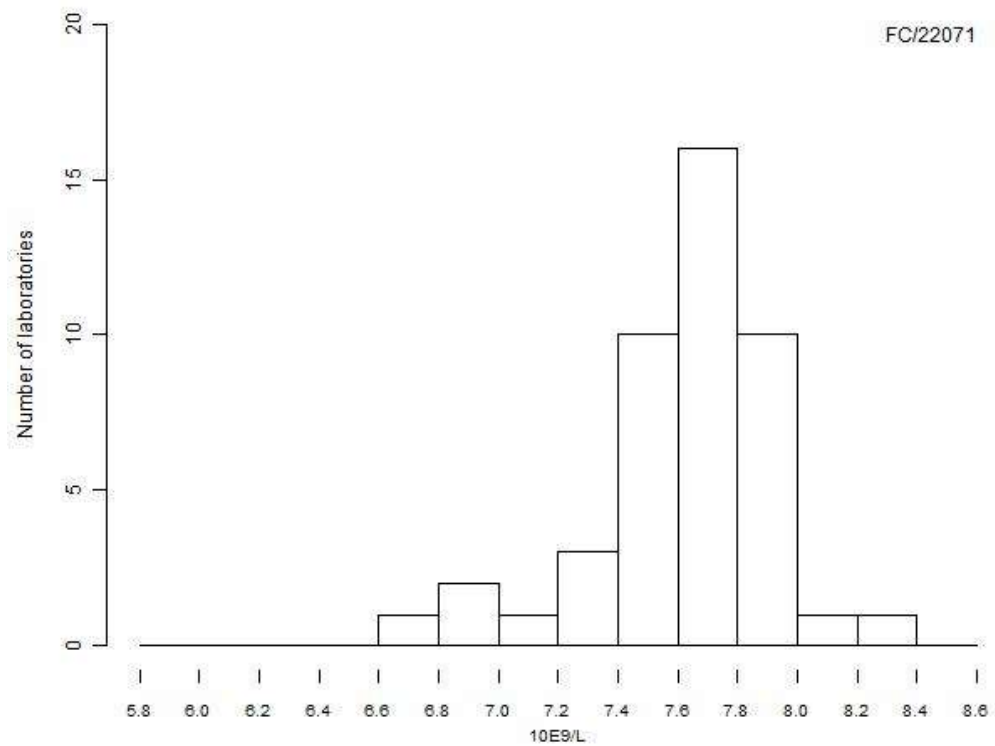
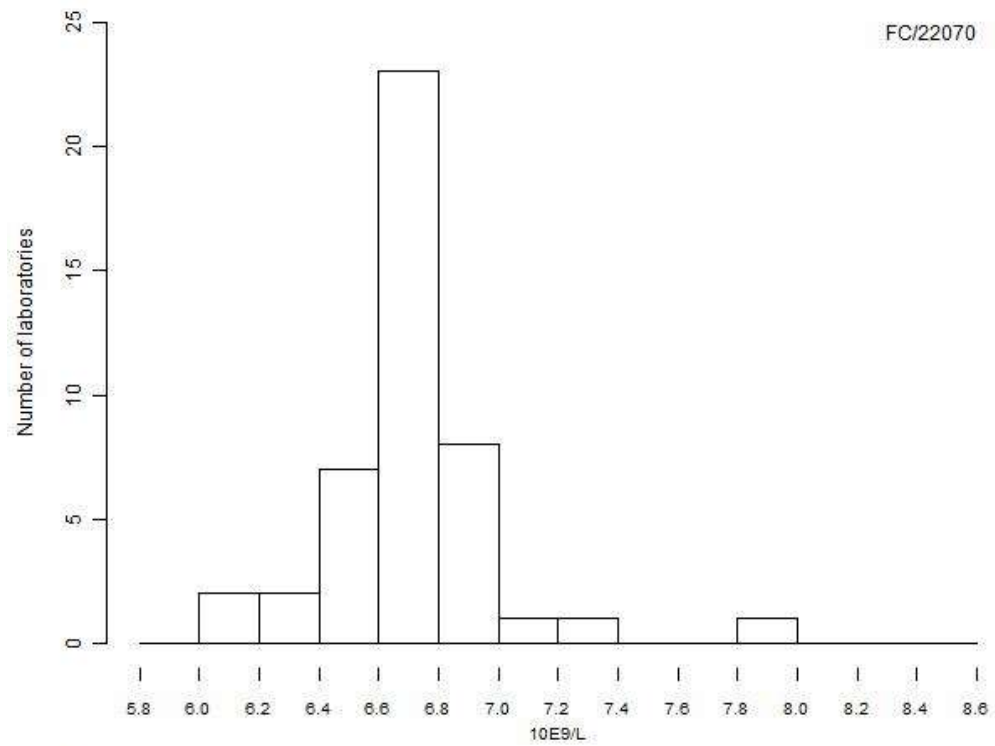
In terms of processing timelines, 21% of laboratories performed the analyses on Day 1, 48% on Day 2, 27% on Day 3, and the remaining 4% on Day 4.

The following tables show the medians and coefficients of variation obtained by the Belgian clinical laboratories for the samples FC/22070 and FC/22071:

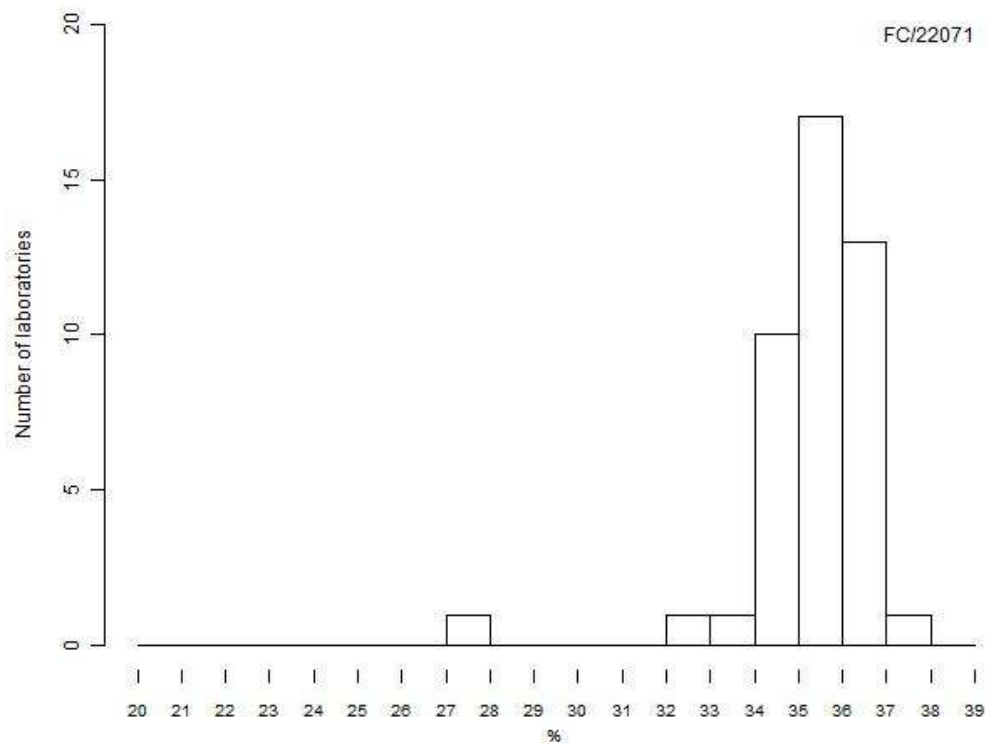
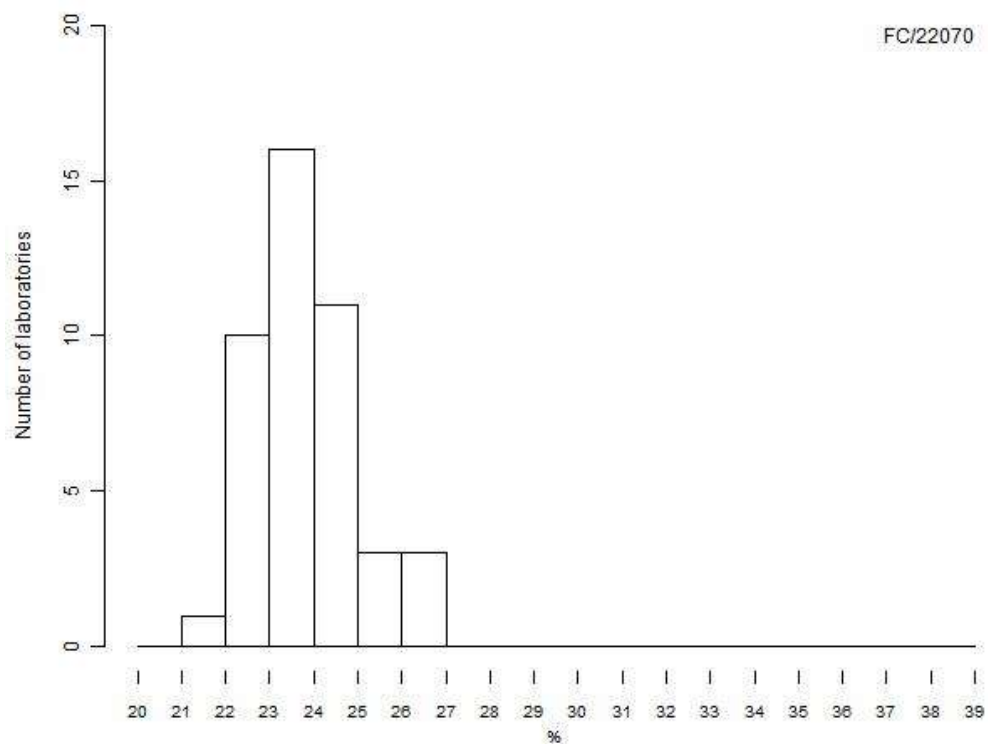
FC/22070	Median	SD	CV,%	N
WBC 10E9/L	6.72	0.16	2.3	47
Lympho% haematology analyser	23.8	1.0	4.0	45
Lympho% flow cytometer	22.7	2.7	12.1	42
CD3 %	75.8	3.0	4.0	49
CD3 10E9/L	1.210	0.113	9.4	49
CD4 %	41.4	3.7	8.9	49
CD4 10E9/L	0.655	0.080	12.2	49
CD8 %	32.1	2.7	8.3	49
CD8 10E9/L	0.528	0.051	9.7	49
CD19 %	19.8	2.4	12.0	48
CD19 10E9/L	0.318	0.055	17.2	48
NKcells %	3.2	0.8	25.4	48
NKcells 10E9/L	0.053	0.016	30.7	48
Kappa % B lymphocytes	62.6	8.8	14.1	40
Lambda % B lymphocytes	36.7	8.1	22.2	40
Kappa/lambda	1.70	0.73	43.2	40
Sum K+L % B lymphocytes	99.9	0.4	0.4	40
Lymphosum %	99.5	0.5	0.5	48

FC/22071	Median	SD	CV,%	N
WBC 10E9/L	7.67	0.21	2.8	47
Lympho% haematology analyser	35.8	1.1	3.1	45
Lympho% flow cytometer	35.4	6.4	18.0	42
CD3 %	68.6	1.9	2.7	49
CD3 10E9/L	1.885	0.105	5.6	49
CD4 %	36.3	2.0	5.5	49
CD4 10E9/L	1.005	0.075	7.4	49
CD8 %	30.4	1.3	4.4	49
CD8 10E9/L	0.855	0.073	8.5	49
CD19 %	11.4	1.3	11.7	48
CD19 10E9/L	0.314	0.042	13.4	48
NKcells %	18.7	2.2	11.9	48
NKcells 10E9/L	0.520	0.077	14.8	48
Kappa % B lymphocytes	61.9	3.8	6.2	40
Lambda % B lymphocytes	37.9	3.9	10.4	40
Kappa/lambda	1.63	0.27	16.6	40
Sum K+L % B lymphocytes	100.0	0.1	0.1	40
Lymphosum %	99.0	2.4	2.5	48

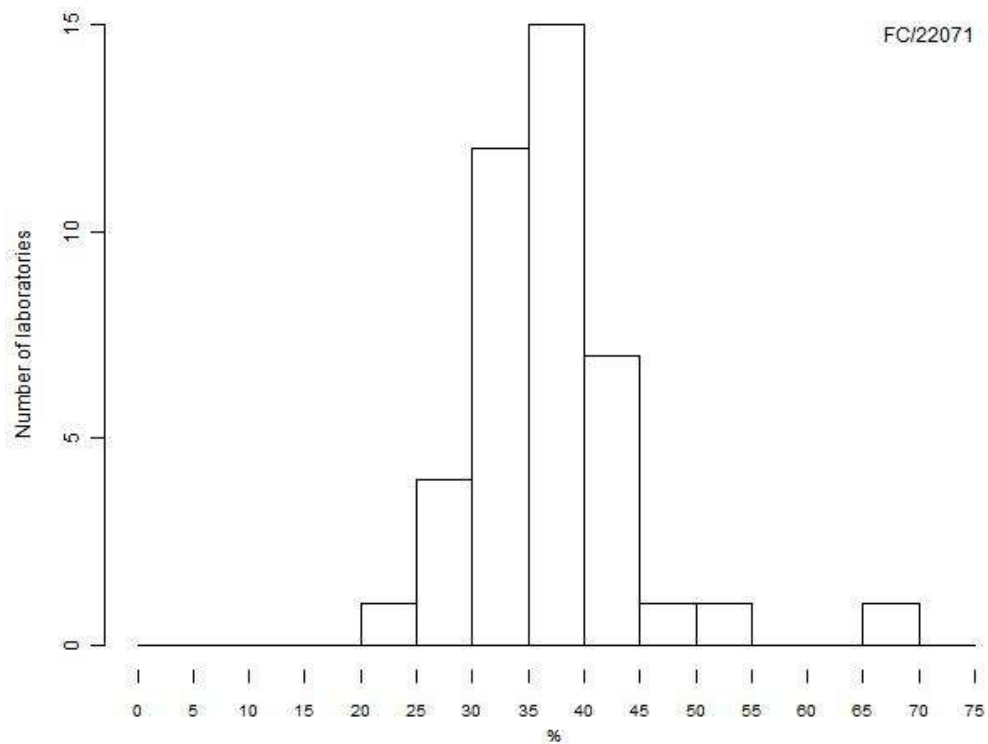
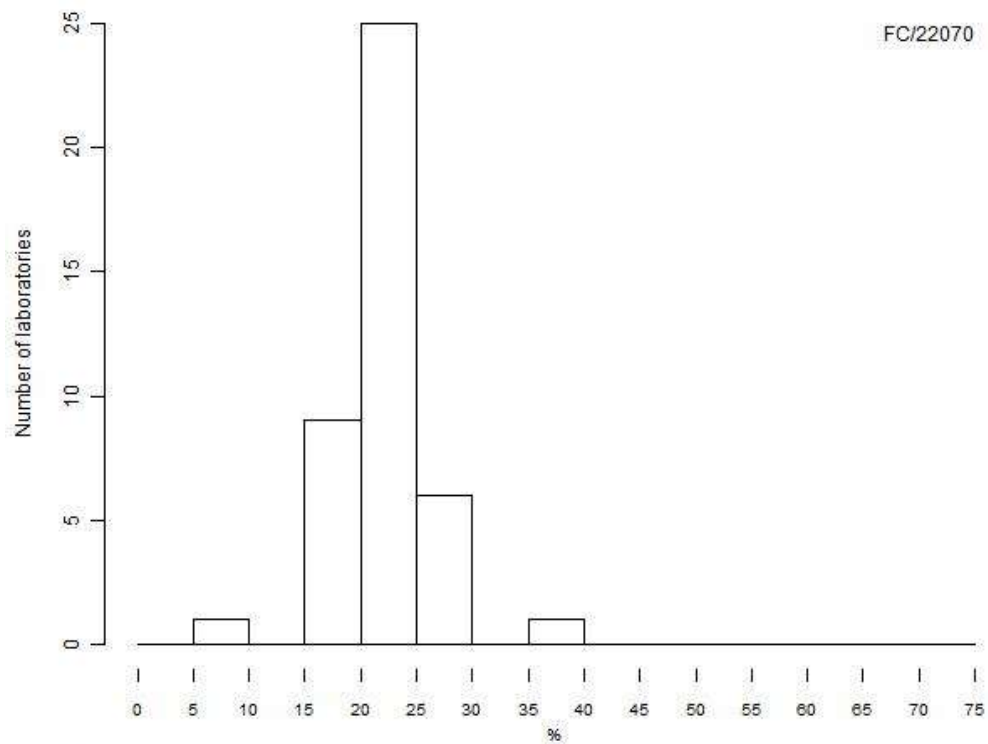
WBC 10E9/L



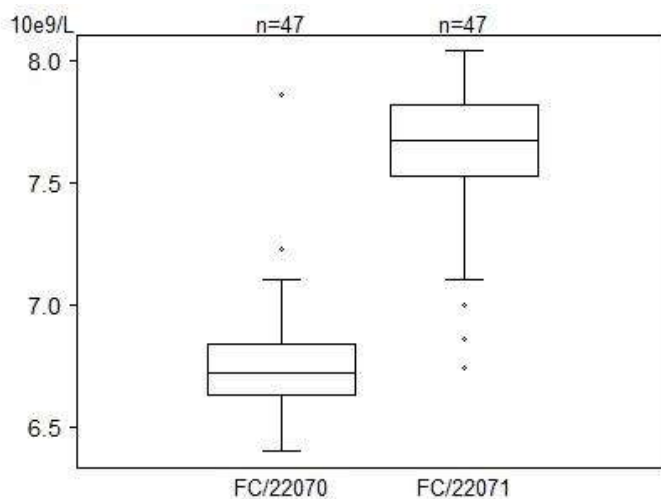
Lympho% haematology analyser



Lympho% flow cytometer



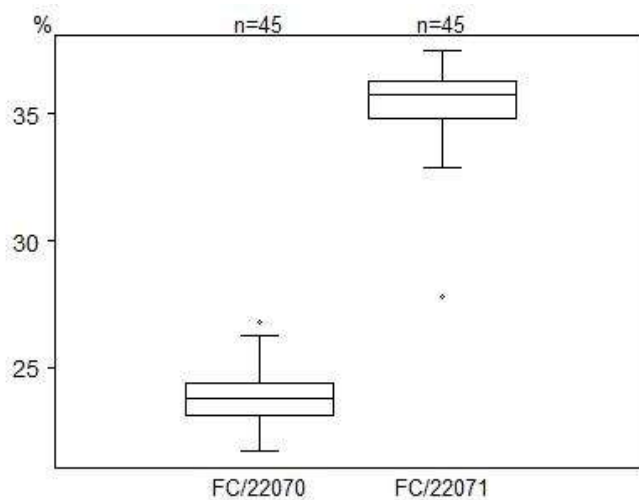
WBC 10E9/L



Results not represented on the graph

FC/22070 = 6.17 10e9/L
 FC/22070 = 6.2 10e9/L
 FC/22070 = 6.28 10e9/L
 FC/22070 = 6650 10e9/L
 FC/22070 = 6720 10e9/L
 FC/22071 = 8.37 10e9/L
 FC/22071 = 7360 10e9/L
 FC/22071 = 7680 10e9/L

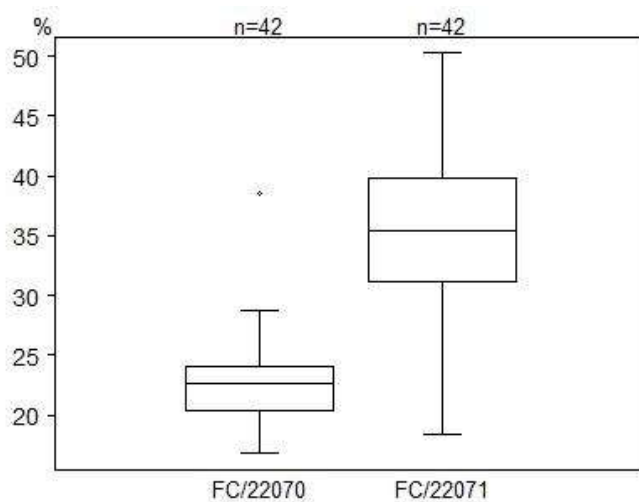
Lympho% haematology analyser



Results not represented on the graph

FC/22070 = 53.6 %
 FC/22071 = 51.3 %

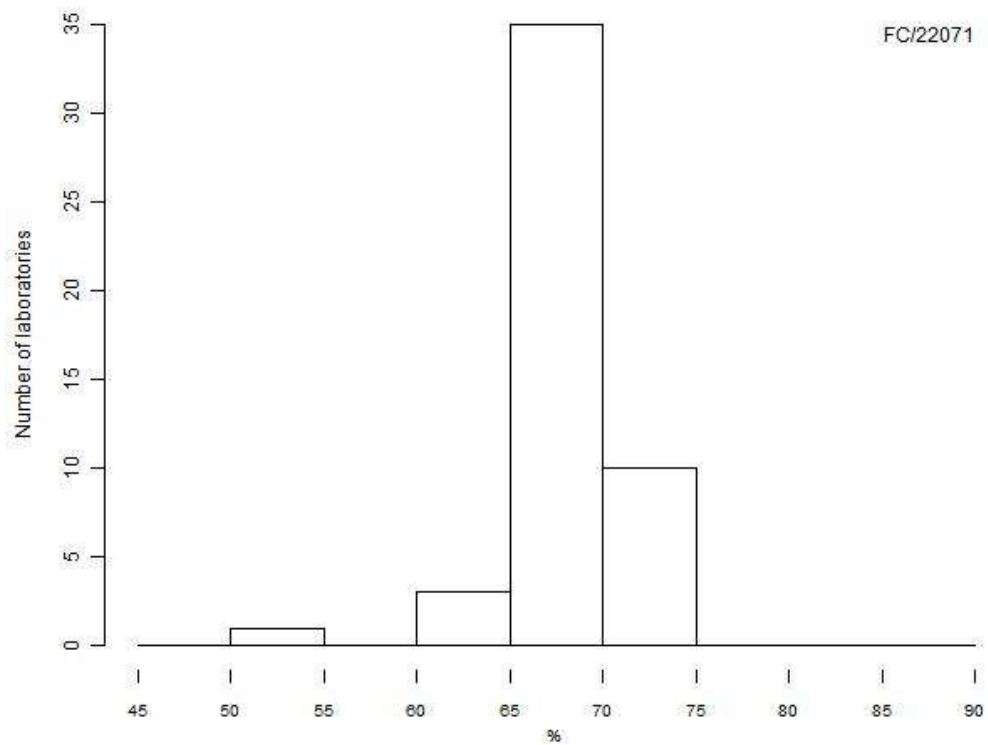
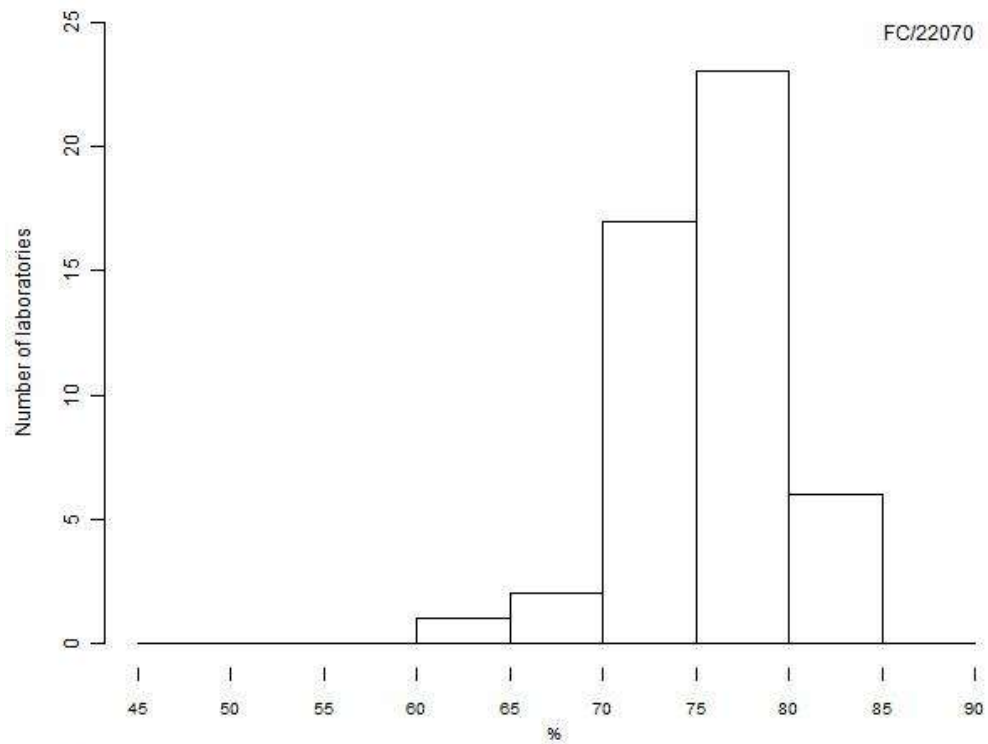
Lympho% flow cytometer



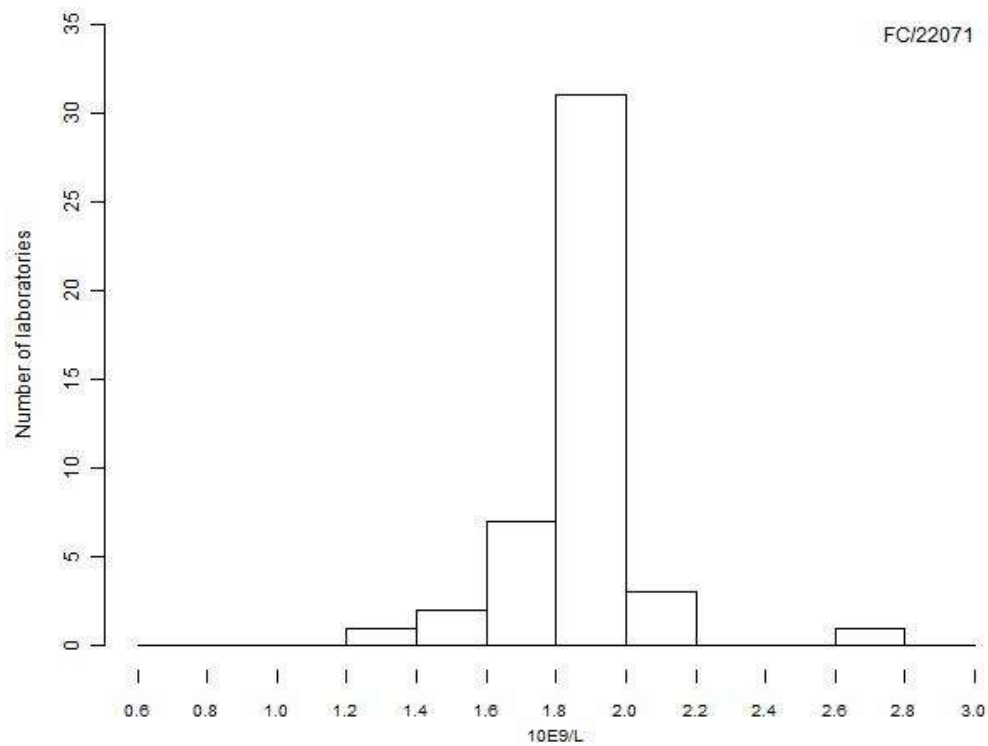
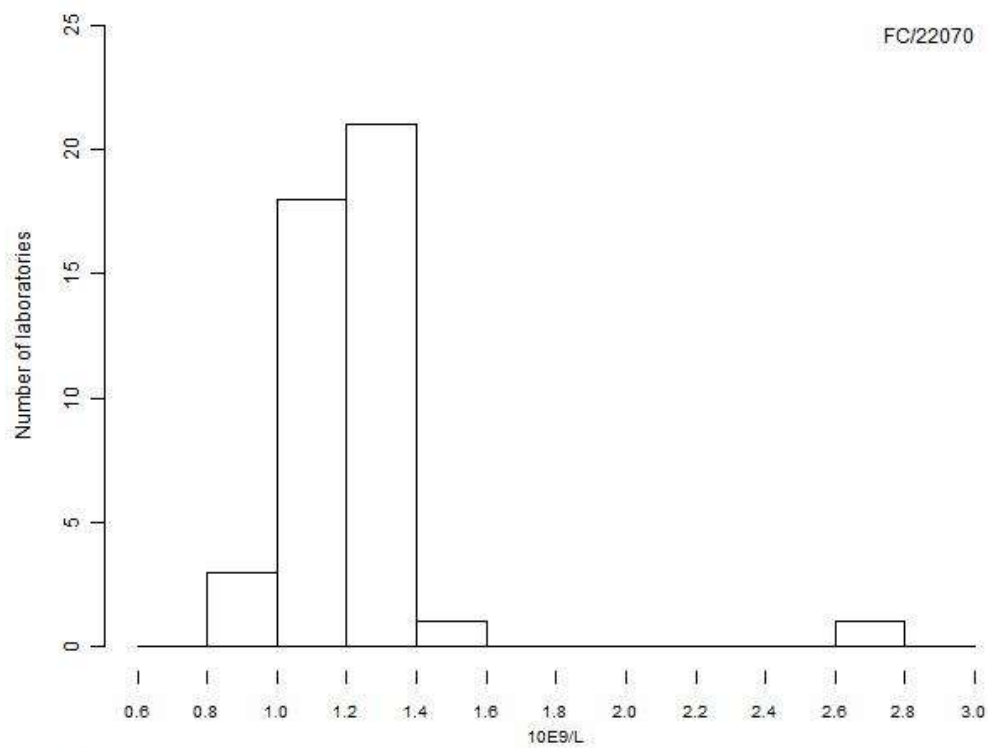
Results not represented on the graph

FC/22070 = 7.4 %
 FC/22071 = 67.5 %

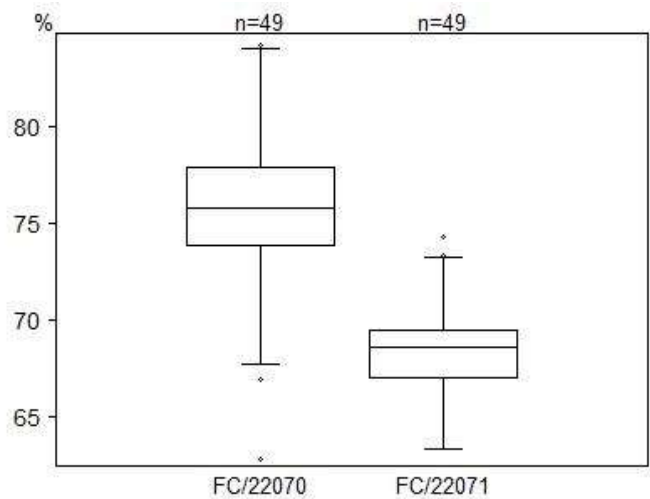
CD3 %



CD3 10E9/L

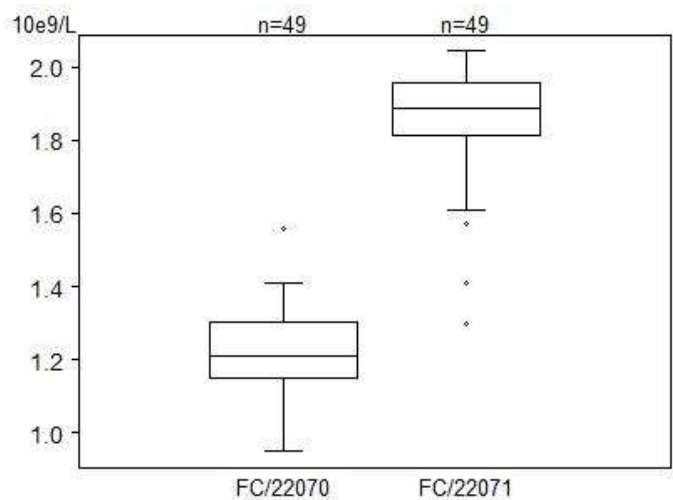


CD3 %



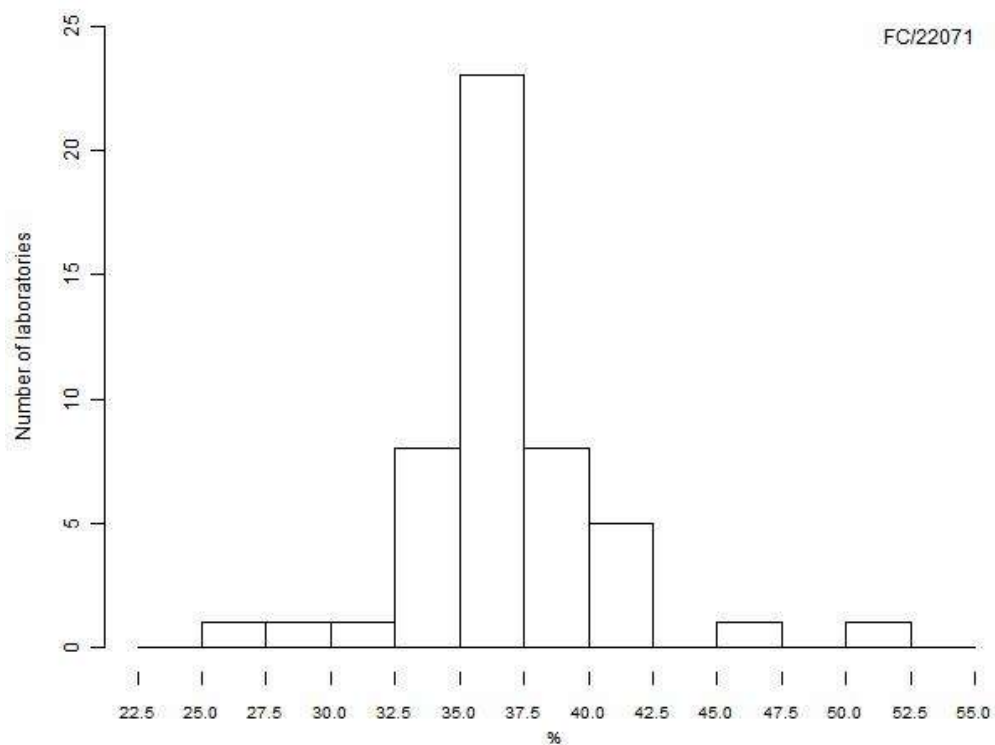
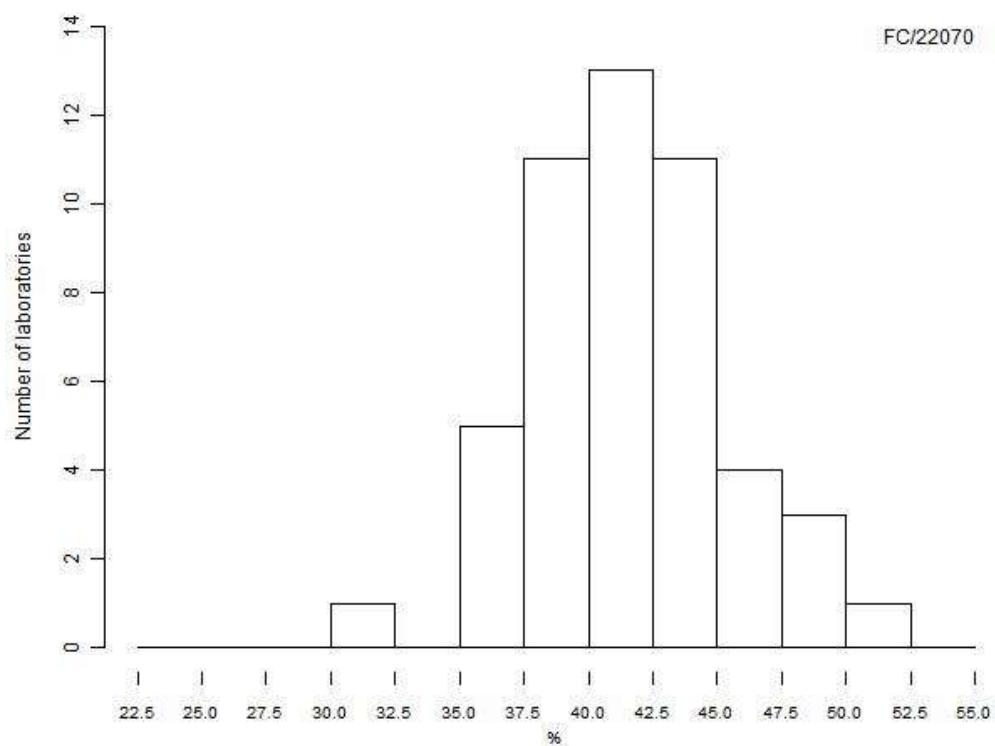
Results not represented
on the graph
FC/22071 = 51.8 %
FC/22071 = 62.2 %

CD3 10E9/L

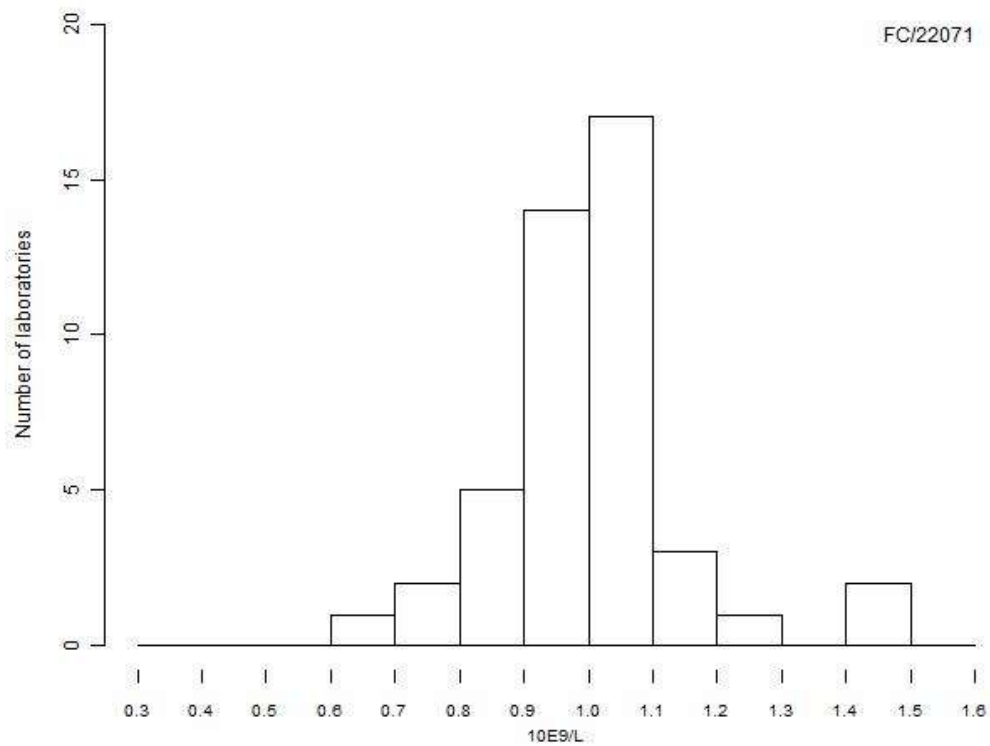
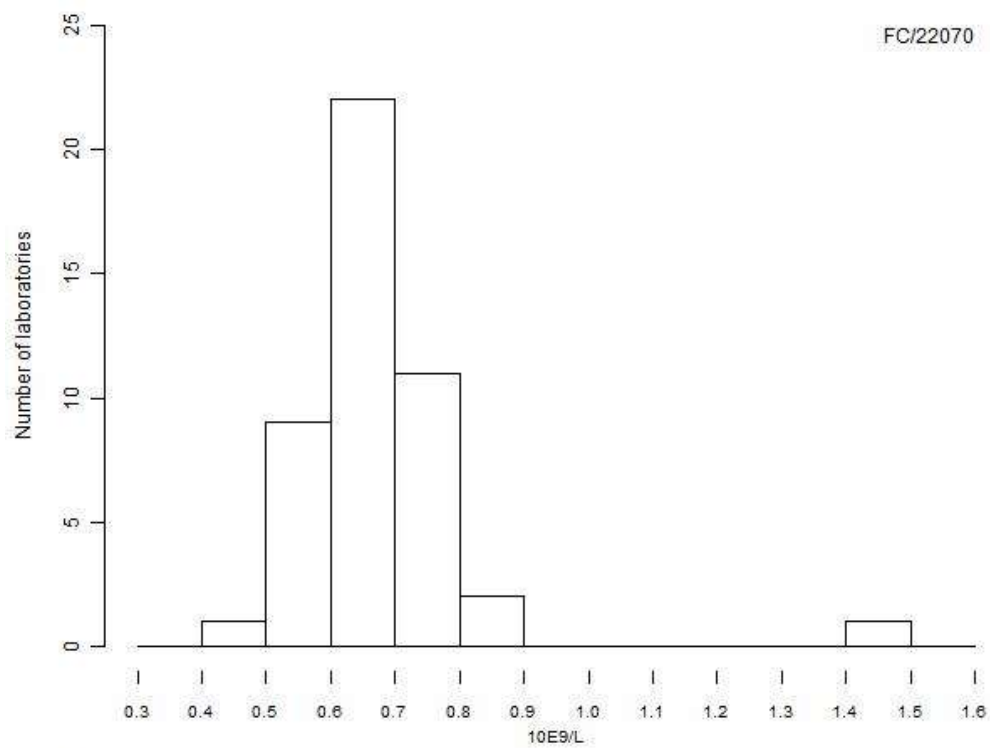


Results not represented
on the graph
FC/22070 = 2.7 10e9/L
FC/22070 = 5.38 10e9/L
FC/22070 = 1148.8 10e
FC/22070 = 1191 10e9/
FC/22070 = 1212 10e9/
FC/22070 = 1278 10e9/
FC/22071 = 2.69 10e9/L
FC/22071 = 5.26 10e9/L
FC/22071 = 1770 10e9/
FC/22071 = 1883.3 10e
FC/22071 = 1998 10e9/

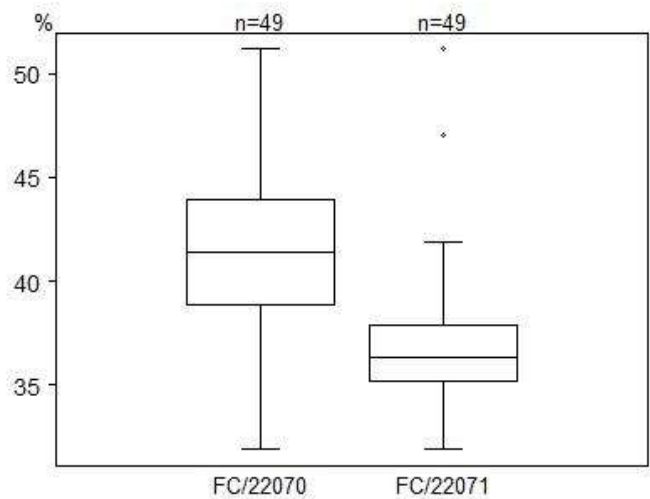
CD4 %



CD4 10E9/L



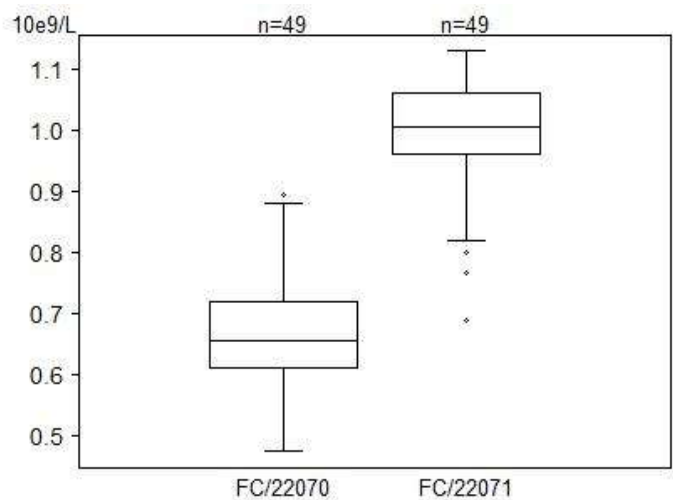
CD4 %



Results not represented
on the graph

- FC/22070 = 30.4 %
- FC/22070 = 52.4 %
- FC/22071 = 27.5 %
- FC/22071 = 29.6 %

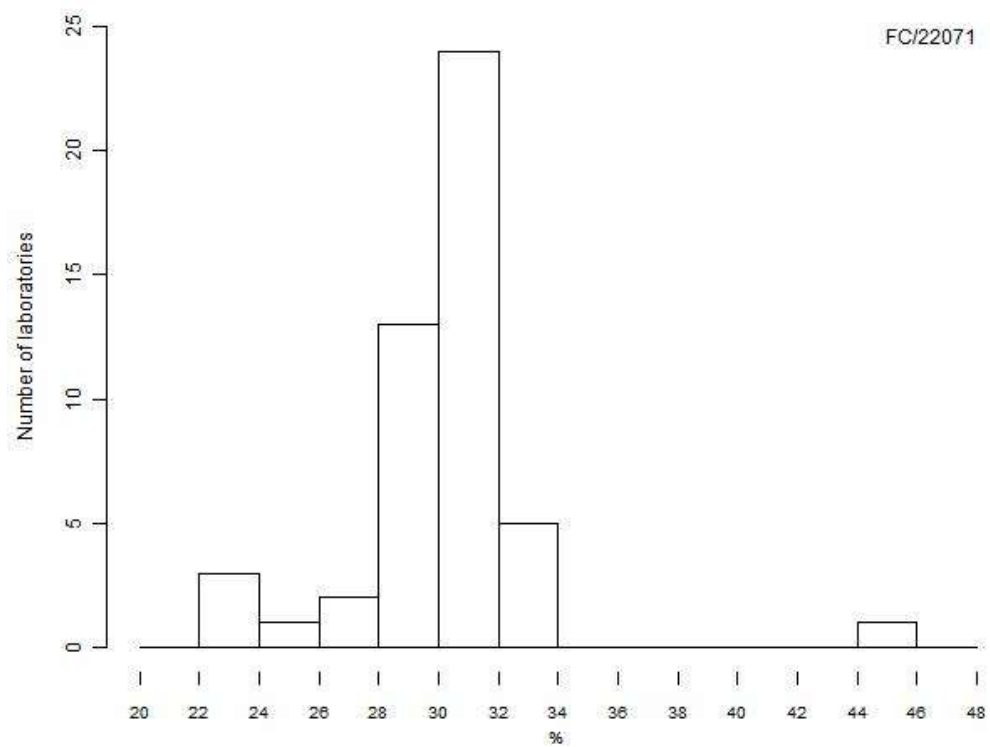
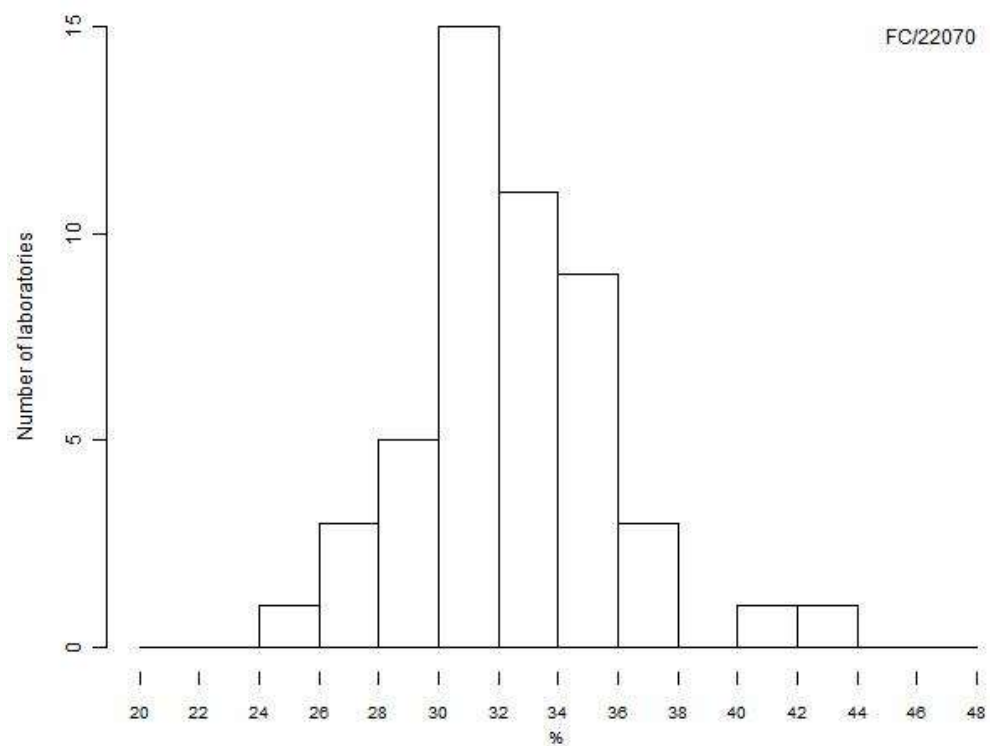
CD4 10E9/L



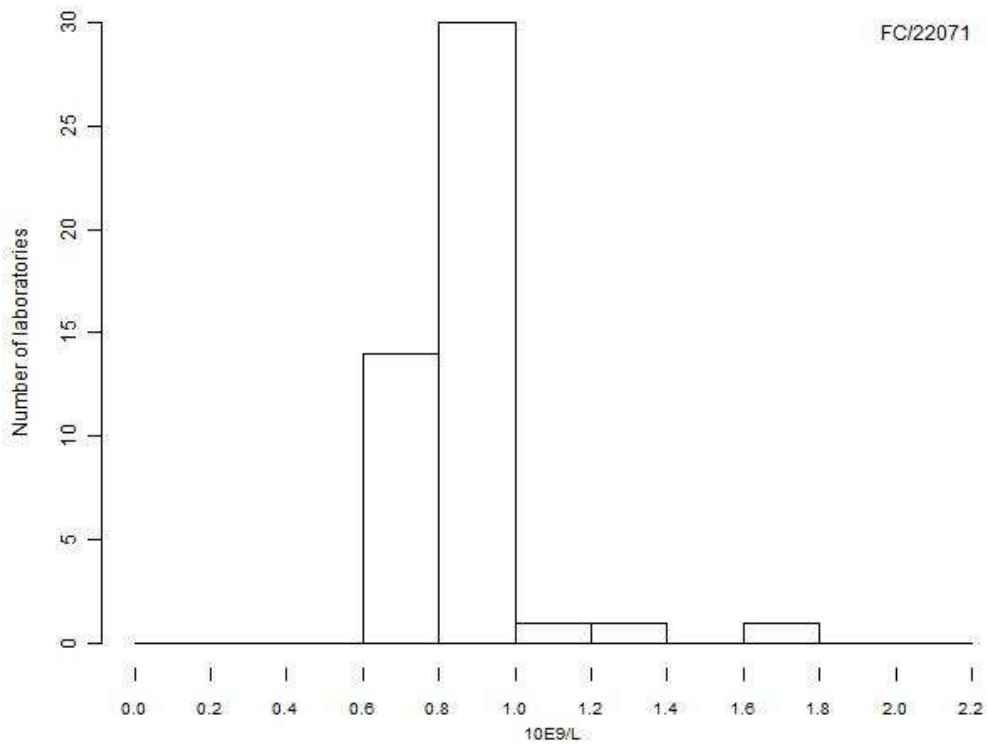
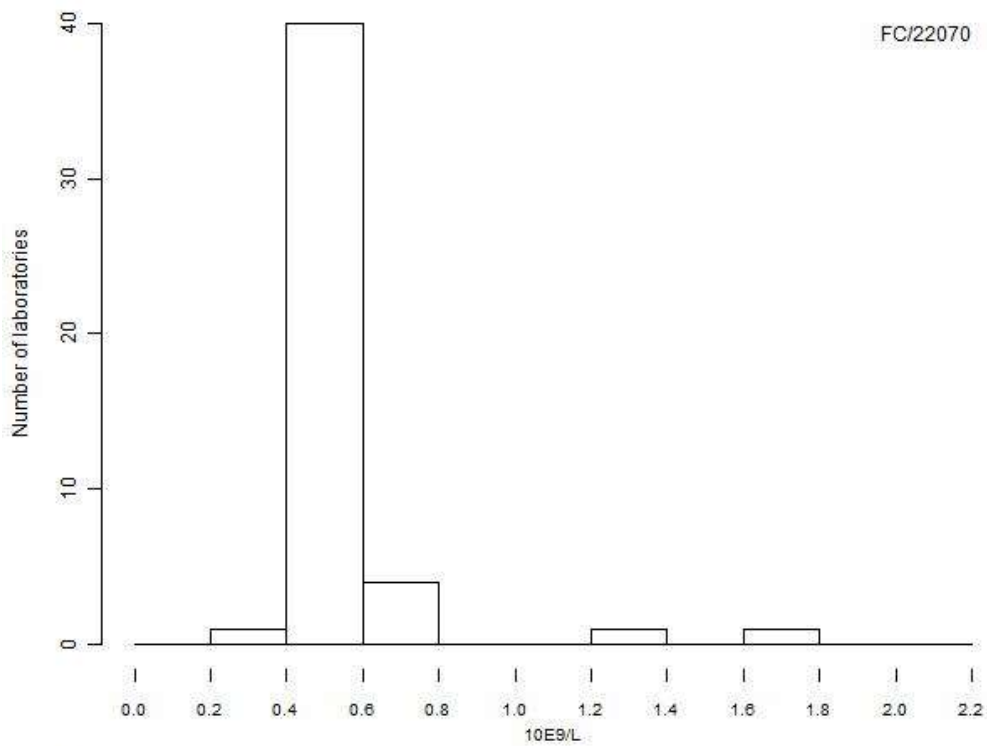
Results not represented
on the graph

- FC/22070 = 1.48 10e9/L
- FC/22070 = 3.34 10e9/L
- FC/22070 = 658.6 10e9
- FC/22070 = 763 10e9/L
- FC/22071 = 1.239 10e9
- FC/22071 = 1.43 10e9/L
- FC/22071 = 1.442 10e9
- FC/22071 = 3.25 10e9/L
- FC/22071 = 981 10e9/L
- FC/22071 = 1003.2 10e
- FC/22071 = 1123 10e9/

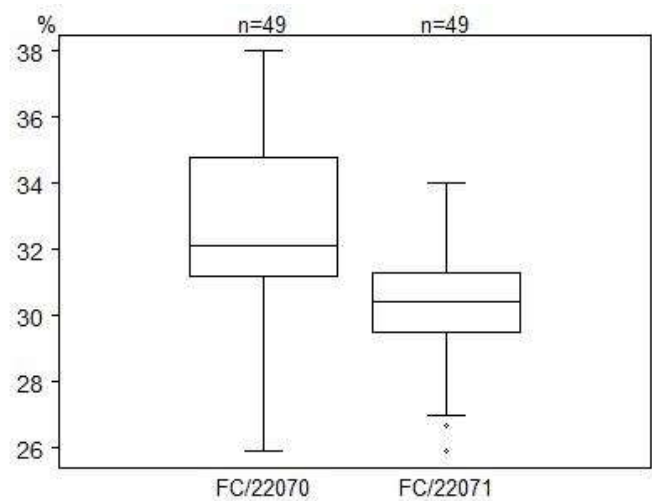
CD8 %



CD8 10E9/L



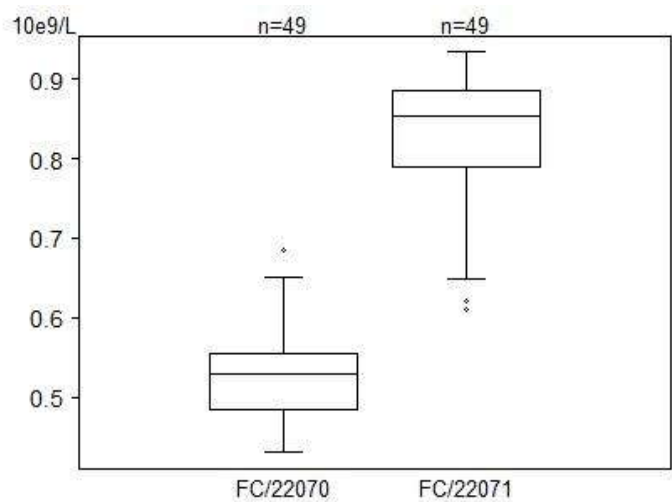
CD8 %



Results not represented
on the graph

- FC/22070 = 24.7 %
- FC/22070 = 41.4 %
- FC/22070 = 43.1 %
- FC/22071 = 22.8 %
- FC/22071 = 23.1 %
- FC/22071 = 24 %
- FC/22071 = 45.3 %

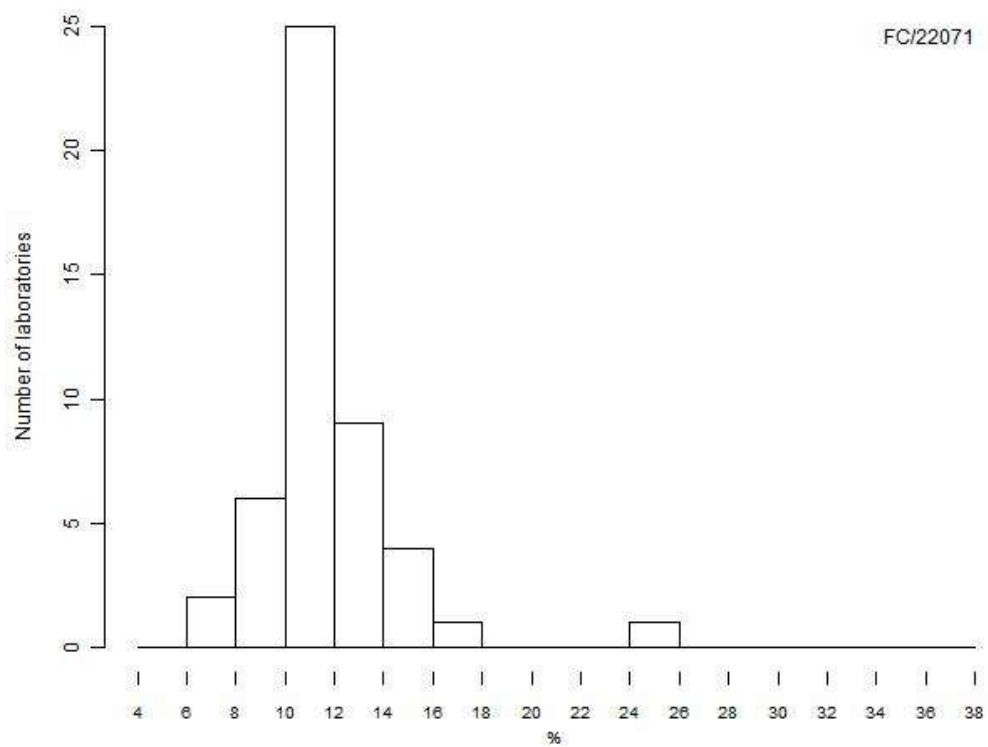
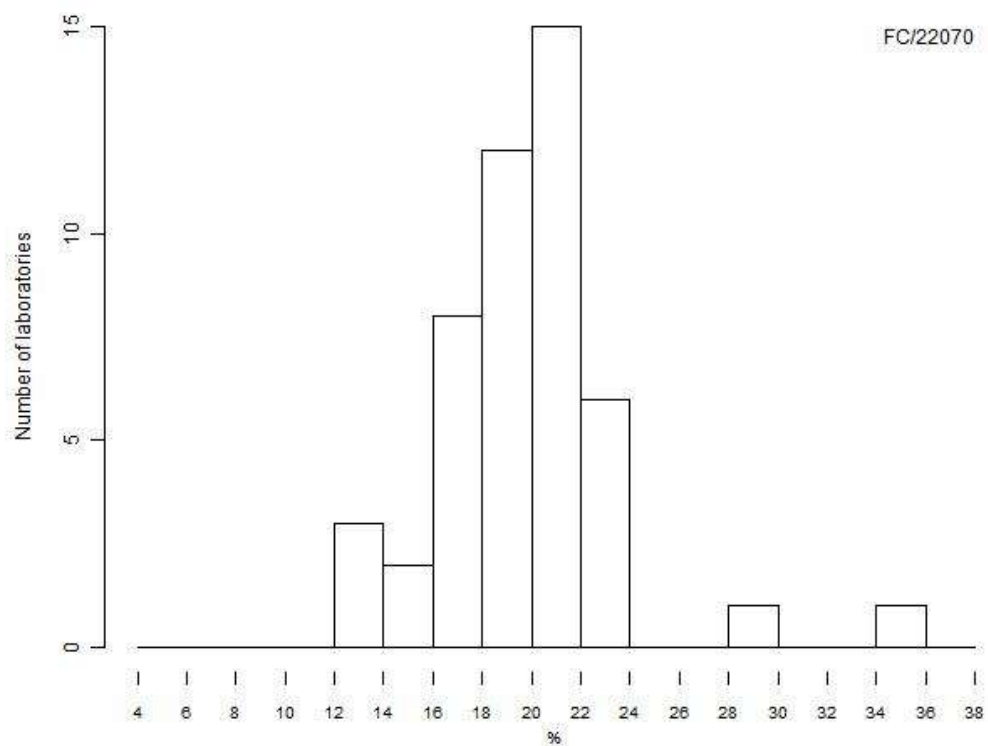
CD8 10E9/L



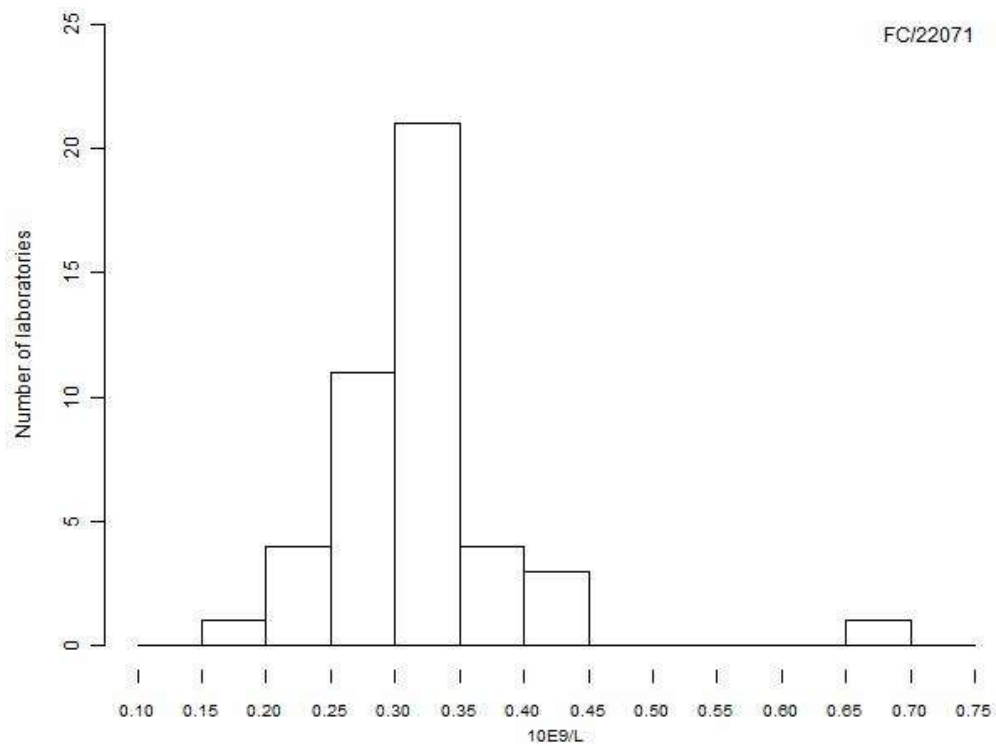
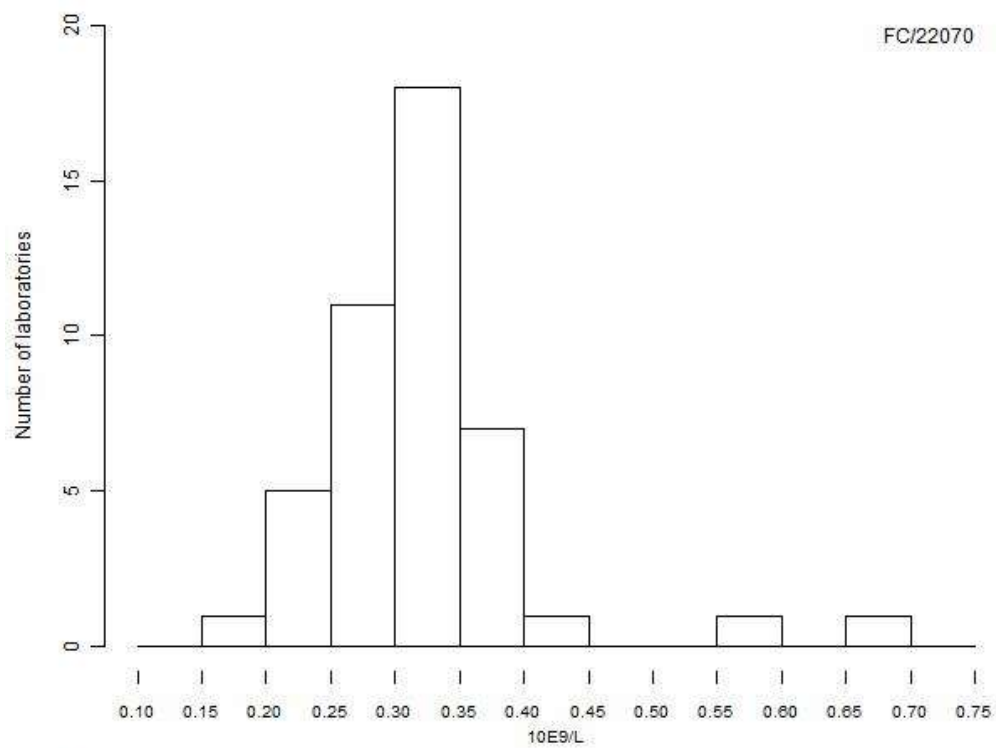
Results not represented
on the graph

- FC/22070 = 0.34 10e9/L
- FC/22070 = 1.21 10e9/L
- FC/22070 = 1.67 10e9/L
- FC/22070 = 428 10e9/L
- FC/22070 = 475.3 10e9
- FC/22071 = 1.18 10e9/L
- FC/22071 = 1.276 10e9
- FC/22071 = 1.8 10e9/L
- FC/22071 = 753 10e9/L
- FC/22071 = 817.7 10e9

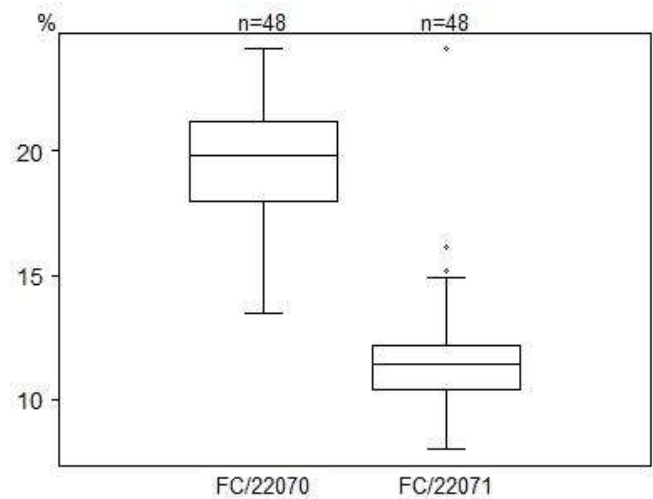
CD19 %



CD19 10E9/L

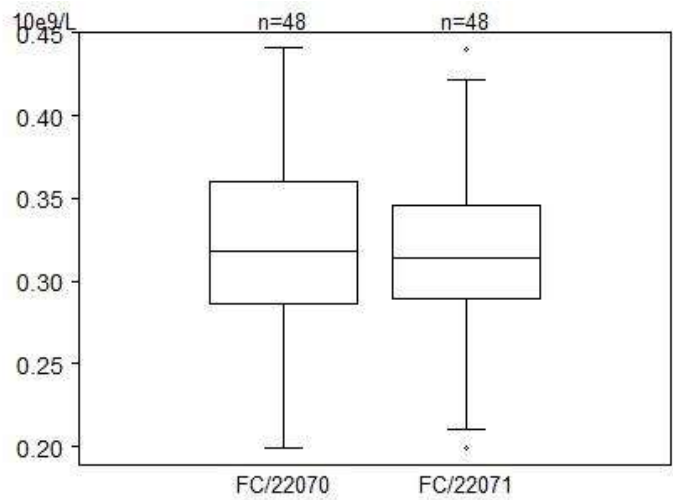


CD19 %



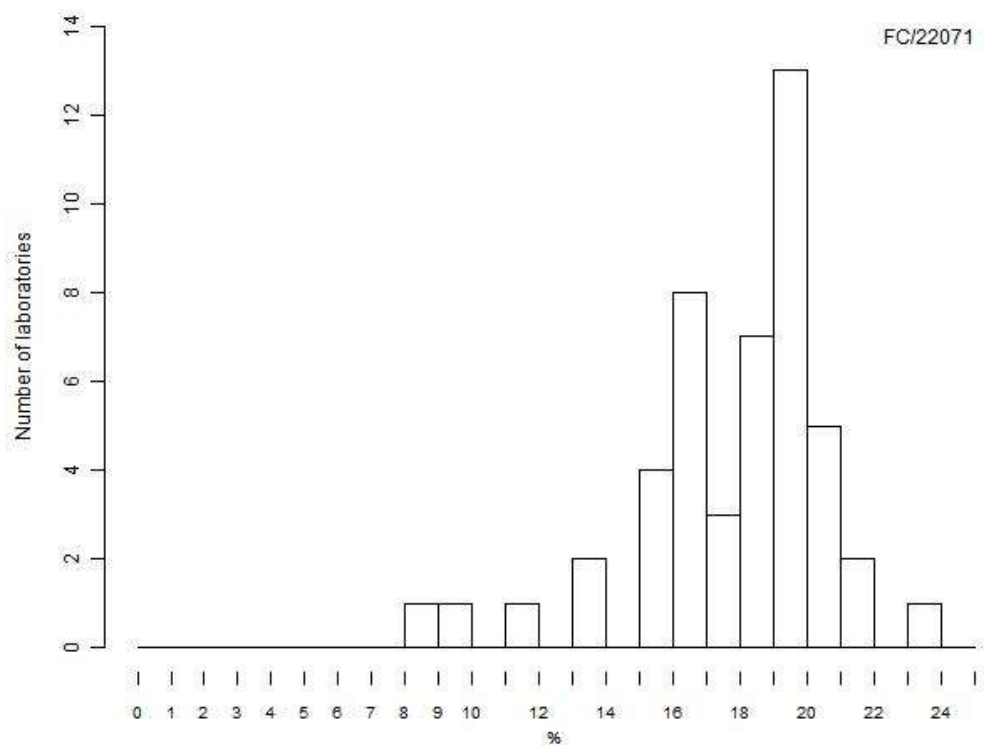
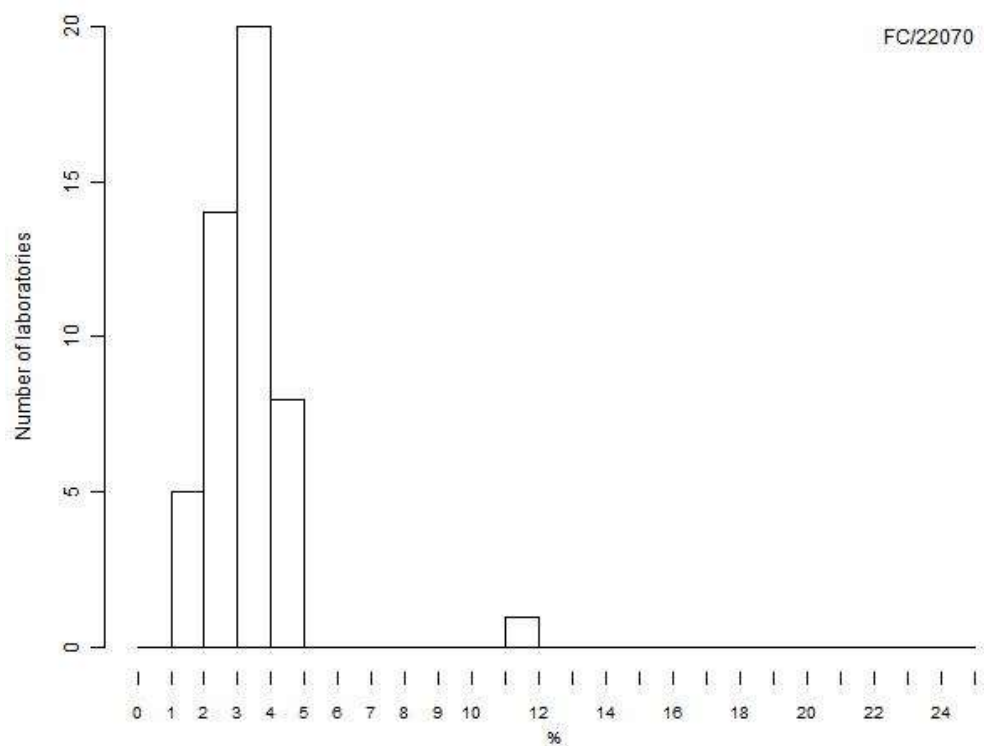
Results not represented
on the graph
FC/22070 = 28.3 %
FC/22070 = 34.2 %

CD19 10E9/L

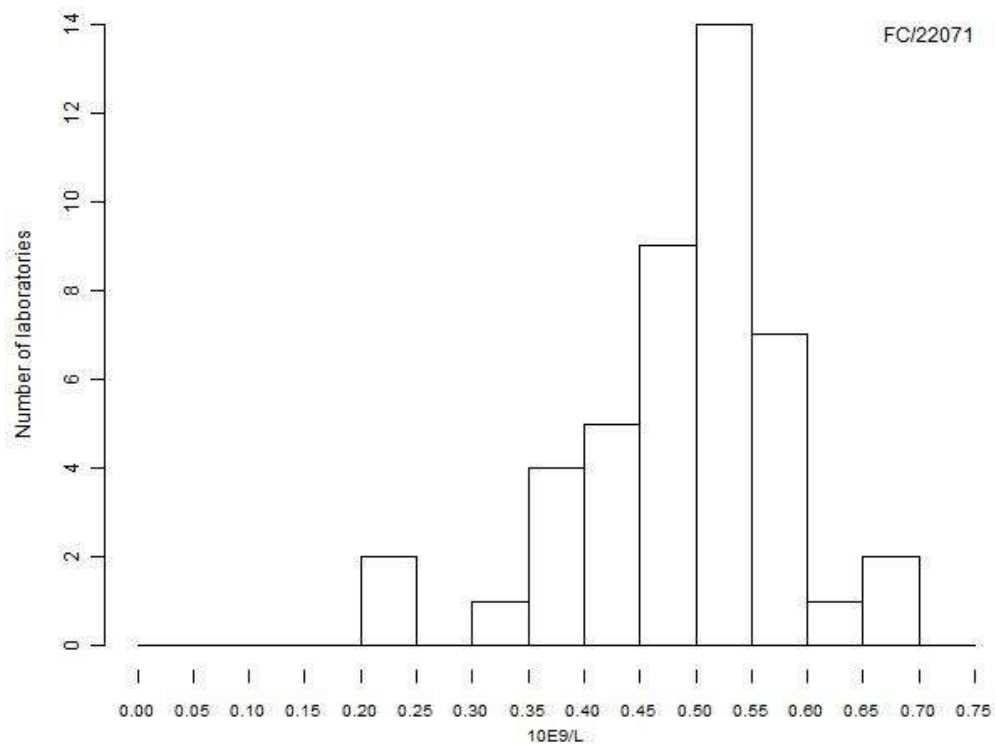
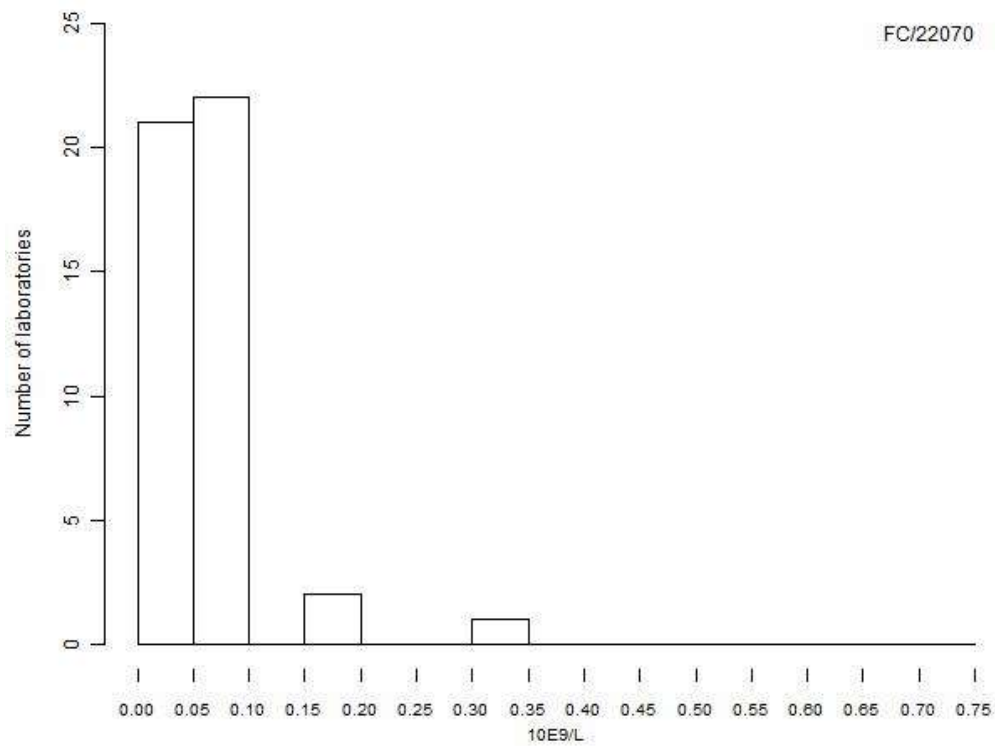


Results not represented
on the graph
FC/22070 = 0.17 10e9/L
FC/22070 = 0.6 10e9/L
FC/22070 = 0.66 10e9/L
FC/22070 = 1.13 10e9/L
FC/22070 = 268.2 10e9
FC/22070 = 343 10e9/L
FC/22071 = 0.66 10e9/L
FC/22071 = 0.96 10e9/L
FC/22071 = 309.1 10e9
FC/22071 = 417 10e9/L

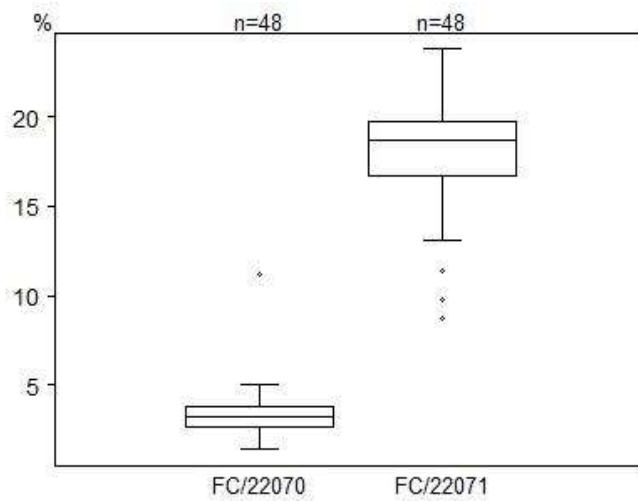
NKcells %



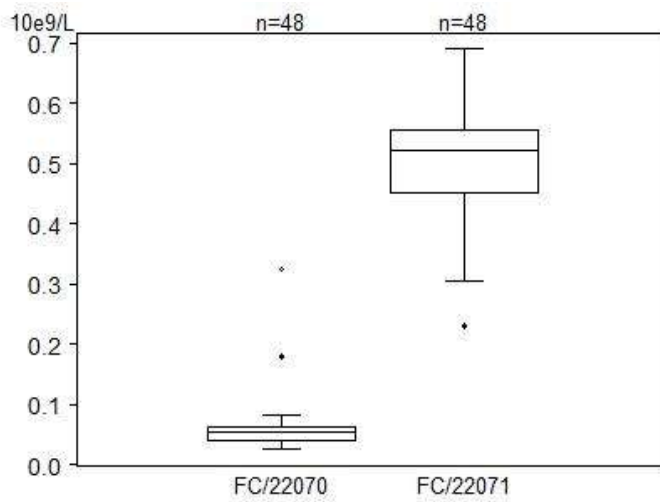
NKcells 10E9/L



NKcells %



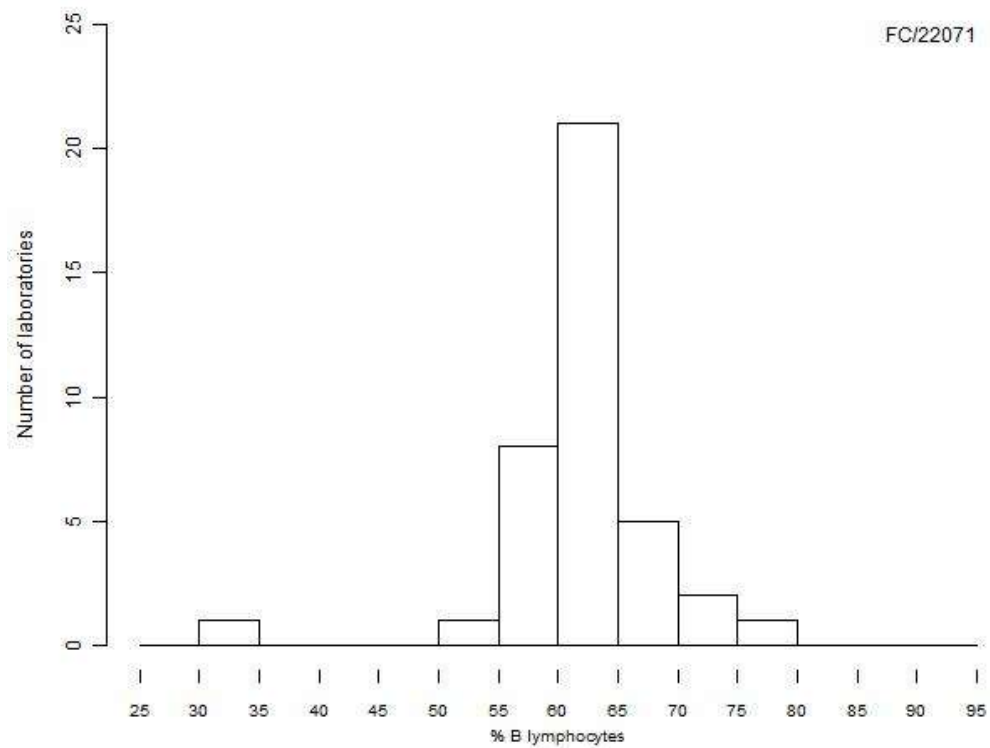
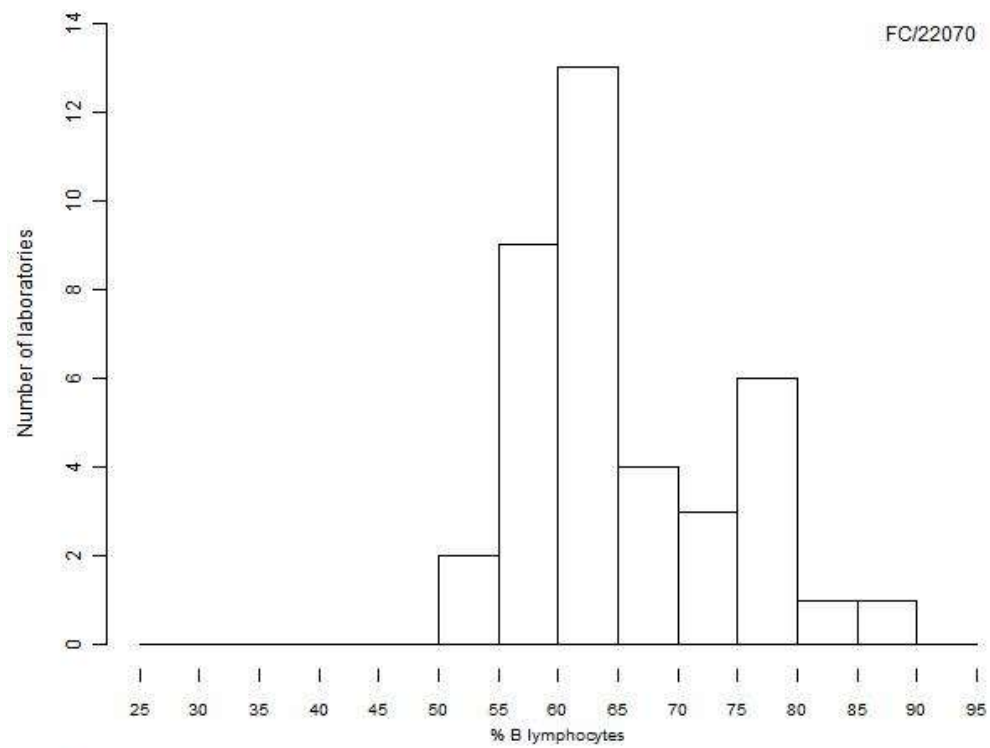
NKcells 10E9/L



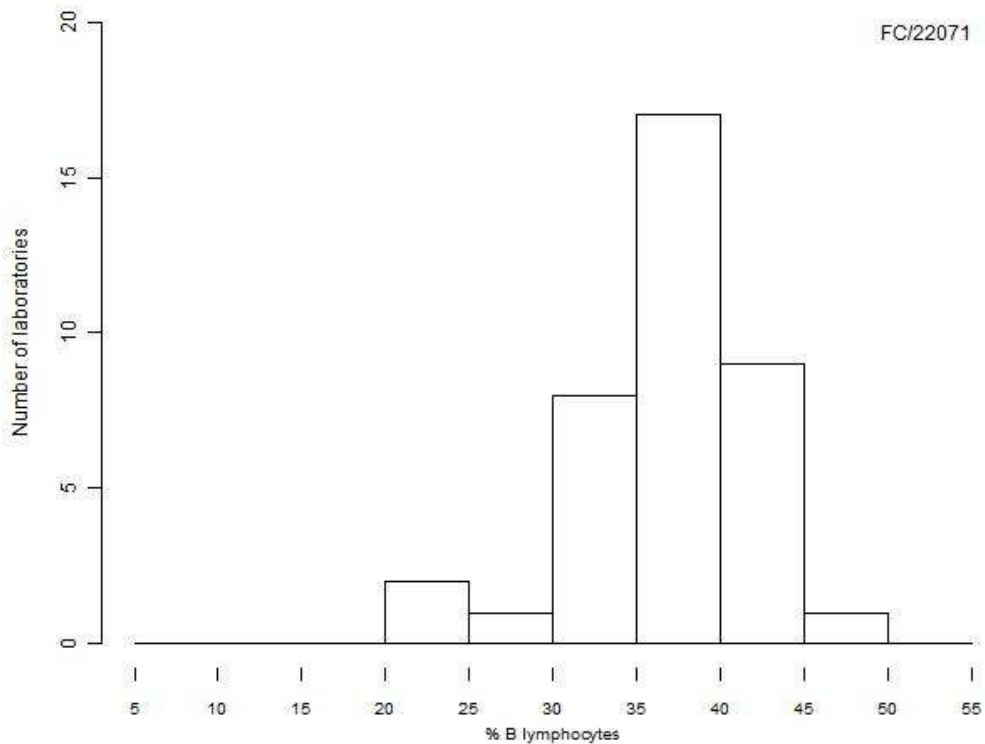
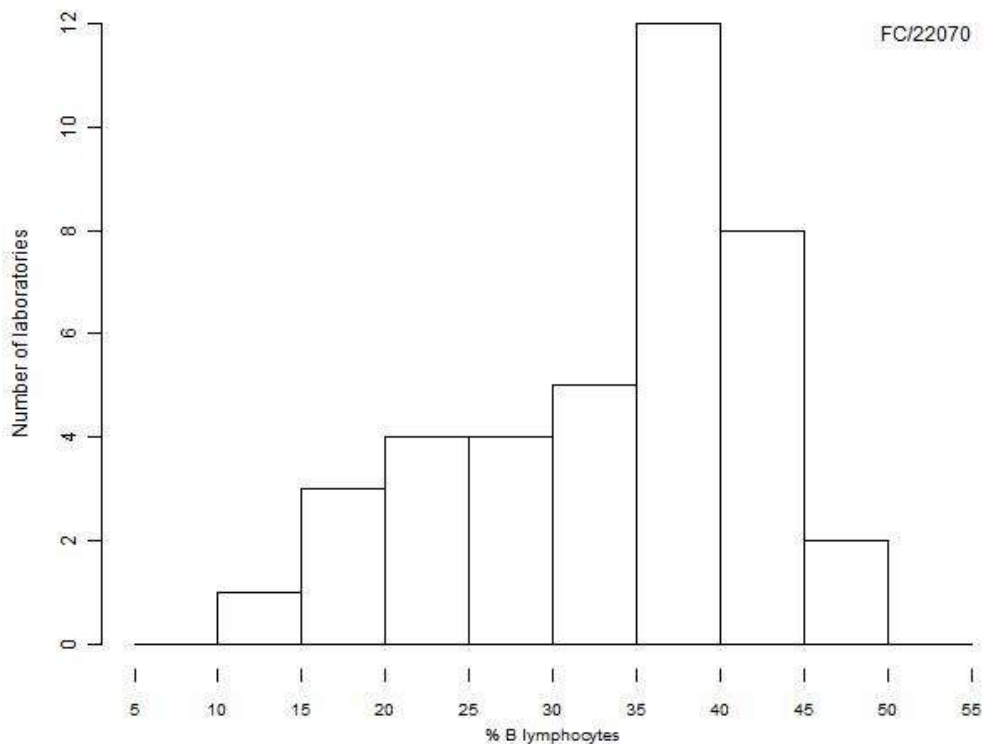
Results not represented
on the graph

FC/22070 = 20.9 10e9/l
FC/22070 = 48 10e9/L
FC/22071 = 1.52 10e9/l
FC/22071 = 414 10e9/L
FC/22071 = 474.9 10e9

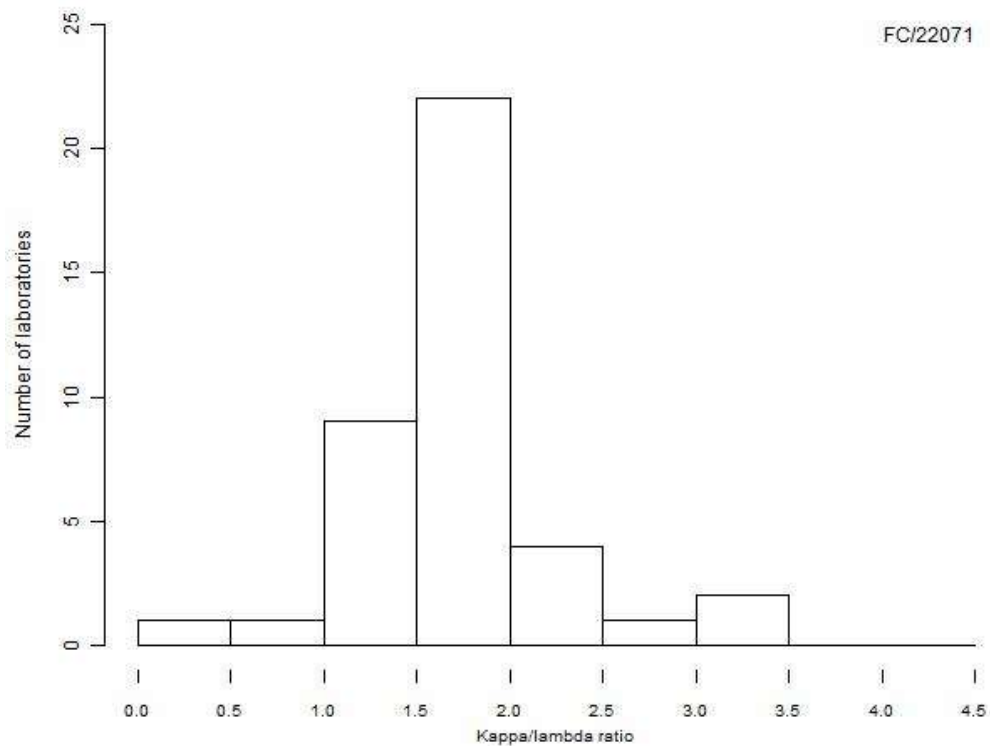
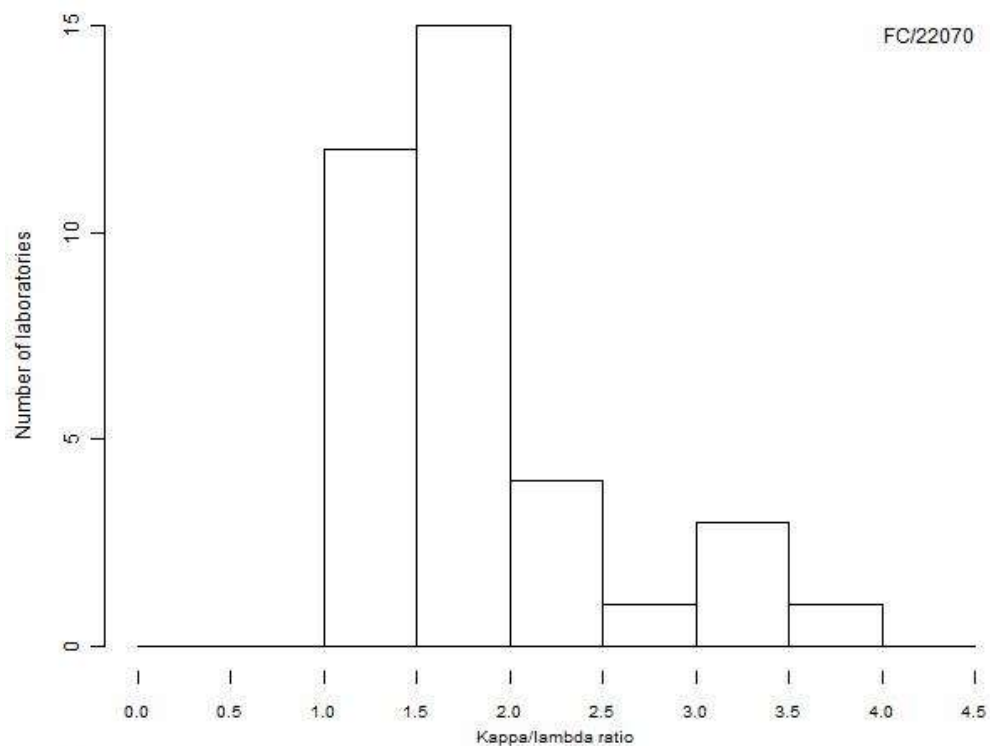
Kappa % B lymphocytes



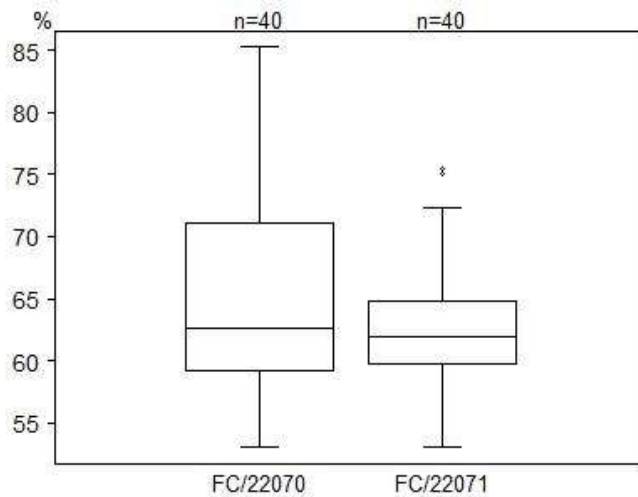
Lambda % B lymphocytes



Kappa/lambda



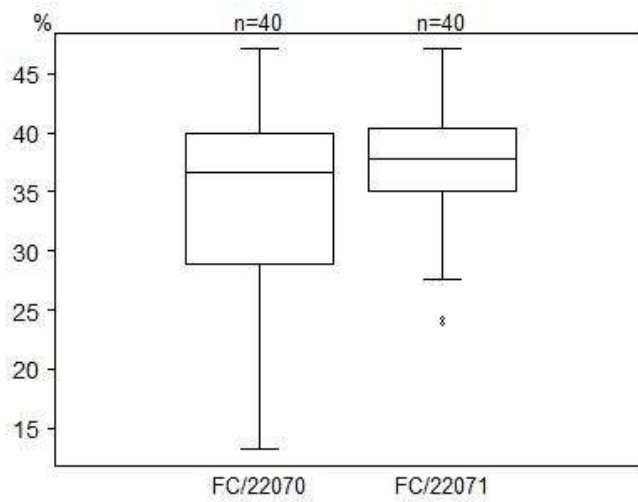
Kappa % B lymphocytes



Results not represented
on the graph

FC/22070 = 8 %
FC/22071 = 4 %
FC/22071 = 32.9 %

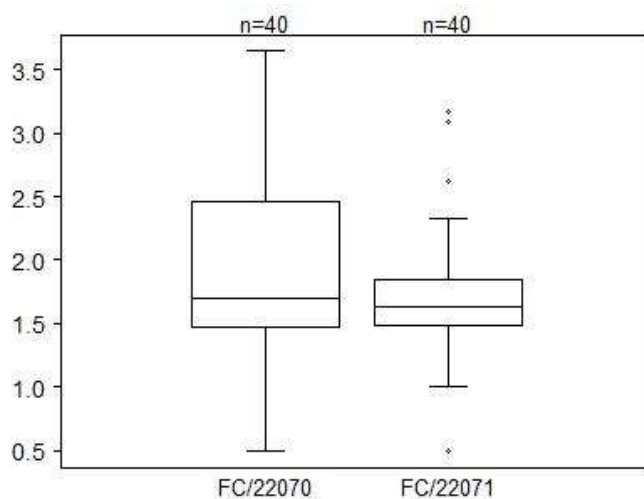
Lambda % B lymphocytes



Results not represented
on the graph

FC/22070 = 6 %
FC/22071 = 4 %
FC/22071 = 67.3 %

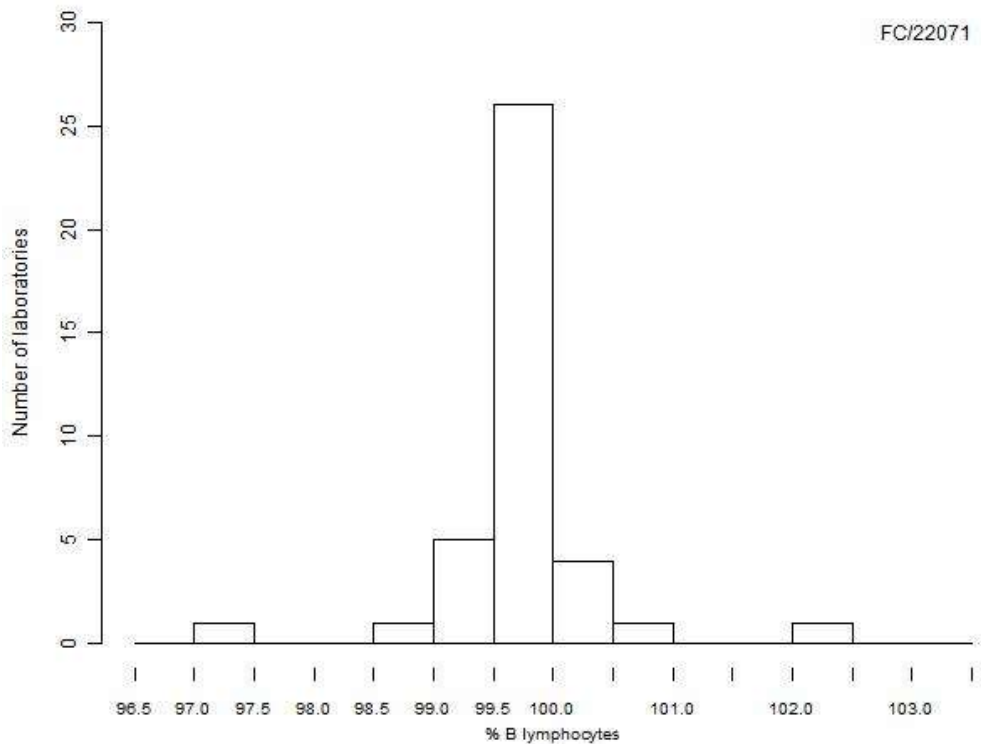
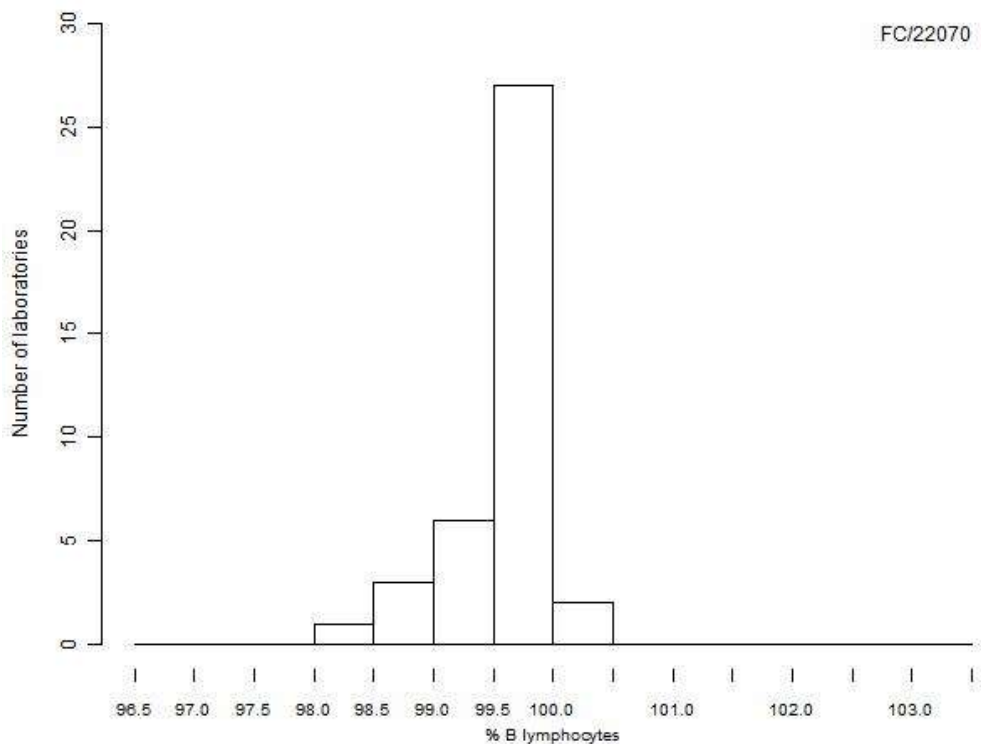
Kappa/lambda



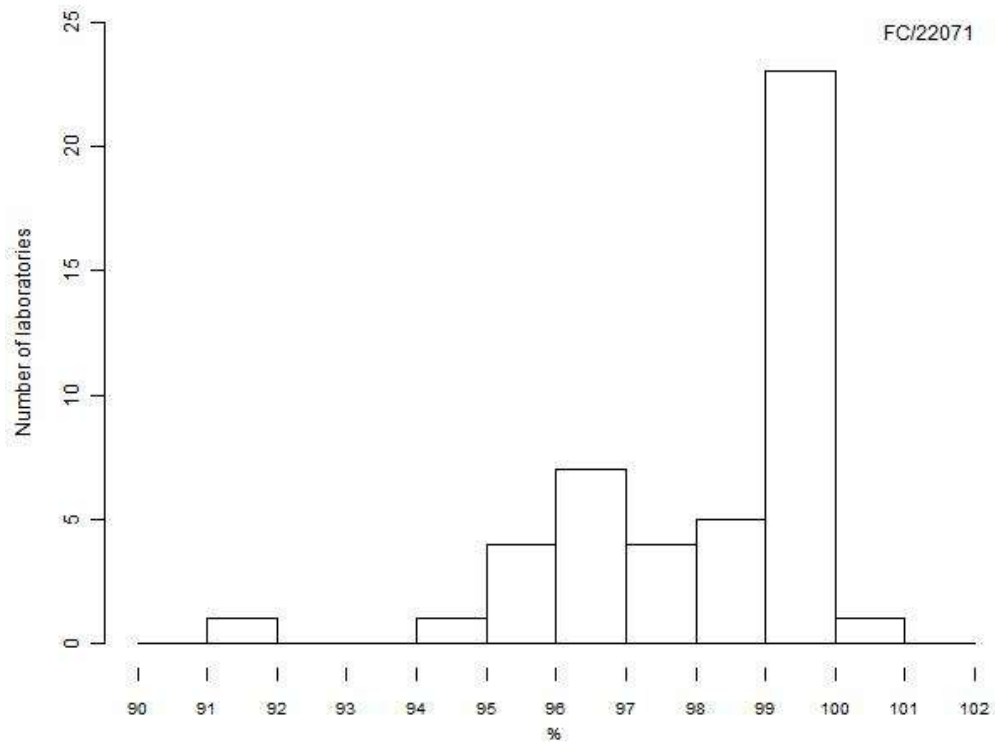
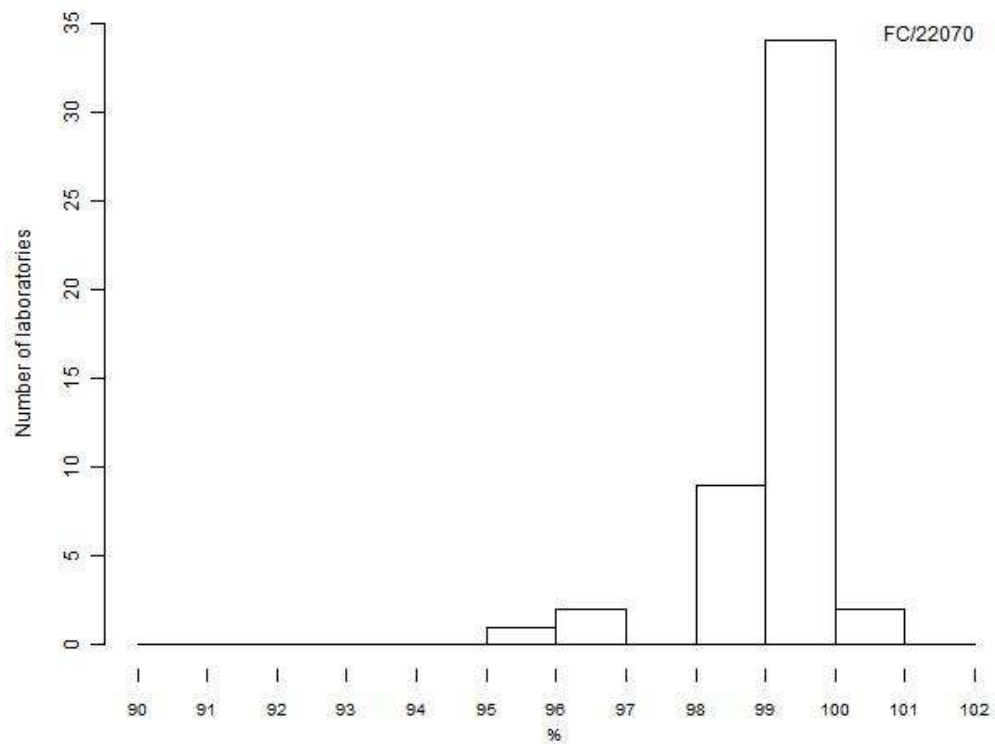
Results not represented
on the graph

FC/22070 = 4.01
FC/22070 = 4.06
FC/22070 = 4.15
FC/22070 = 6.46

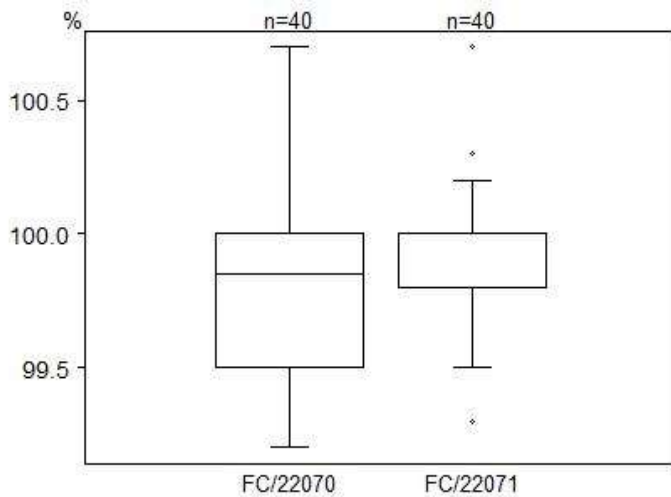
Sum K+L % B lymphocytes



Lymphosum %



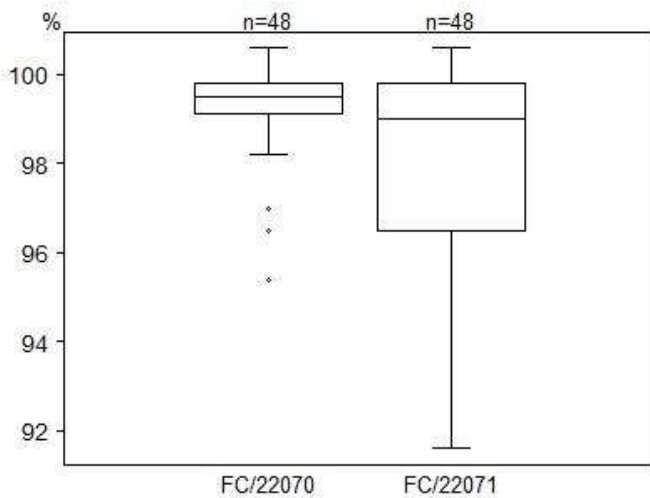
Sum K+L % B lymphocytes



Results not represented
on the graph

FC/22070 = 14 %
FC/22070 = 98.5 %
FC/22070 = 98.6 %
FC/22070 = 98.6 %
FC/22070 = 98.7 %
FC/22071 = 8 %
FC/22071 = 97.4 %
FC/22071 = 98.6 %
FC/22071 = 102.5 %

Lymphosum %



Results not represented
on the graph

FC/22071 = 88.7 %
FC/22071 = 89 %

For technical validation purposes it is worth noting that in non-pathological peripheral blood of adults the sum of kappa and lambda (expressed as a % of CD19+ B-cells) should be between 90 and 110. The lymphosum (sum of CD3+% plus CD19+% plus CD3⁺CD16⁺ and/or CD56⁺%) should equal the purity of the lymphocytes in the gate $\pm 5\%$, with a maximum variability of $\leq 10\%$.

The table below provides an overview of the percentages of unacceptable results, defined as those with a z-score outside the acceptable limits [-3, 3] for each parameter.

	Percentage of unacceptable results		
	2026/1	2025/3	2025/2
WBC 10E9/L	7%	6%	1%
Lympho% haematology analyser	4%	6%	8%
Lympho% flow cytometer	4%	5%	3%
CD3 %	4%	2%	2%
CD3 10E9/L	5%	4%	9%
CD4 %	4%	4%	4%
CD4 10E9/L	8%	2%	11%
CD8 %	7%	4%	0%
CD8 10E9/L	10%	2%	7%
CD19 %	4%	8%	5%
CD19 10E9/L	5%	4%	3%
NKcells %	4%	3%	0%
NKcells 10E9/L	6%	4%	2%
Kappa % B lymphocytes	6%	8%	3%
Lambda % B lymphocytes	6%	6%	1%
Kappa/lambda	10%	5%	3%
Sum K+L % B lymphocytes	12%	10%	9%
Lymphosum %	6%	7%	2%

END

© Sciensano, Brussels 2026.

This report may not be reproduced, published or distributed without the consent of Sciensano.

The laboratories' individual results are confidential. They are not passed on by Sciensano to third parties, nor to members of the Commission, the Committees of experts or the working group EQA.